

Log-concavity and Approximate Counting for Totally Unimodular Polytopes

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Abstract

We present a new lower bound on the number of lattice points of all totally unimodular polytopes, generalizing previous lower bounds on contingency tables, integer flows, and beyond. Our bound is based on the Gurvits capacity convex optimization problem, and thus our result implies an efficient deterministic algorithm for approximate counting of the lattice points up to an explicit exponential factor. We achieve our bounds by showing that the associated generating polynomials fit into a new general class of log-concave polynomials called VLC (“variable-wise log-concavity”). This also implies a conjecture of Ferroni and Higashitani on the evaluations of the Ehrhart polynomials of unimodular polytopes. The essential ingredient for these results is the resolution of Barvinok’s log-concavity conjecture for contingency tables on lines, which was proven using ChatGPT 6 Astra. We conjecture a generalization of Barvinok’s conjecture, which we believe will lead to stronger and more general bounds.

1 Introduction

Totally unimodular polytopes are integral polytopes defined by a totally unimodular constraint matrix; concretely, a convex polytope Q is totally unimodular (TU) if there exists a totally unimodular matrix M (i.e., all minors are in $\{-1, 0, +1\}$) and an integral vector $\mathbf{b}$ such that $Q = \{\mathbf{y} \geq 0 : M\mathbf{y} = \mathbf{b}\}$. Such polytopes have a wide range of applications, with their lattice points often representing integer solutions to various combinatorial problems. These applications include job assignments, contingency tables, bipartite matchings, bipartite independent sets, network flows, routing configurations, and beyond.

In this paper, we address two natural questions for such polytopes:

1. How can we estimate the number of lattice points of such a polytope Q ?
2. How does the number of lattice points of Q vary with the choice of $\mathbf{b}$?

Answers to both of these questions arise by proving log-concavity properties of generating functions associated to totally unimodular polytopes.

Given a totally unimodular polytope $Q = \{\mathbf{y} \geq 0 : M\mathbf{y} = \mathbf{b}\}$, there is a natural generating function for its lattice points based on the choice of $\mathbf{b}$. Concretely, if M is $m \times n$ and has columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ then

$$P_M(\mathbf{x}) = \prod_{j=1}^n \frac{1}{1 - \mathbf{x}^{\mathbf{a}_j}}$$

is a generating function with the property that the $\mathbf{x}^{\mathbf{b}}$ coefficient of $P_M(\mathbf{x})$ is precisely the number of lattice points of Q . Both of the above questions then become questions about P_M : how can we estimate the coefficients of P_M , and how can we understand how the coefficients of P_M relate to one another?

To address these questions, we prove that truncations of P_M satisfy a certain natural log-concavity property called VLC. This property says that, after applying coefficient extractions and positive evaluations to all but one variable of P_M , the coefficients of the remaining polynomial are log-concave. This generalizes various notions of log-concave polynomials (strongly log-concave [Gur09], completely log-concave [ALOV24], Lorentzian [BH20], etc.) which have had widespread applications in mathematics and computer science.

The fact that truncations of P_M are VLC is then based on the underlying main technical result of the paper: that the coefficients of P_M are log-concave along any rational line. This says in fact that as $\mathbf{b}$ moves along any line, the number of lattice points of Q forms a log-concave sequence. Barvinok originally asked a more general question of this form for the case of contingency tables [Bar07], and we resolve this line version here for all TU polytopes. This gives a nice answer to question 2, while the full strength of Barvinok’s question for TU polytopes is still open Theorem 1.7.

The VLC property then enables us also to answer question 1. The Gurvits capacity method was originally developed to give a simple proof of the van der Waerden conjecture for permanents in [Gur08]. This technique implies a deterministic simply exponential approximation algorithm for non-negative matrix permanents as well. Since this original work, the capacity method has seen numerous applications to bounding and deterministically approximating various other combinatorial quantities. The general idea is to use log-concavity properties of a given generating function to approximate its coefficients by an associated convex optimization problem. The VLC property is then tailor-made precisely to allow the Gurvits capacity method to work as usual, and thus we can apply this method to P_M and related generating functions.

1.1 Our Results

We now formally answer question 1: how can we estimate the number of lattice points of a TU polytope? We give two results in this direction. The first gives general bounds for all TU polytopes, and the second gives simply exponential bounds in a slightly restricted regime.

Theorem 1.1 (Theorem 4.5). *Let M be an $m \times n$ totally unimodular matrix with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$. For all $\mathbf{b} \in \mathbb{Z}^m$ such that $\#\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n \mid M\mathbf{y} = \mathbf{b}\} < \infty$, we have*

$$\mathcal{E} \geq \#\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n \mid M\mathbf{y} = \mathbf{b}\} \geq e^{-m}(1 + n\|\mathbf{b}\|_1)^{-m}\mathcal{E},$$

where

$$\mathcal{E} = \inf_{\mathbf{x} > 0} \mathbf{x}^{-\mathbf{b}} \prod_{j=1}^n \frac{1 - (\mathbf{x}^{\mathbf{a}_j})^{\|\mathbf{b}\|_1 + 1}}{1 - \mathbf{x}^{\mathbf{a}_j}} = \inf_{\mathbf{x} > 0} \mathbf{x}^{-\mathbf{b}} \prod_{j=1}^n \sum_{k=0}^{\|\mathbf{b}\|_1} \mathbf{x}^{k\mathbf{a}_j}$$

is a convex program up to log-log transformation.

Corollary 1.2 (Theorem 4.6). *Let M be an $m \times n$ totally unimodular matrix with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ and $\max\{m, n\} \geq 3$. For all $\mathbf{b} \in \mathbb{Z}^m$ satisfying $\|\mathbf{b}\|_1 \geq 6m \cdot \log(\max\{m, n\})$ and such that $\#\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n \mid M\mathbf{y} = \mathbf{b}\} < \infty$, we have*

$$\mathcal{E} \geq \#\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n \mid M\mathbf{y} = \mathbf{b}\} \geq e^{-\|\mathbf{b}\|_1}\mathcal{E},$$

where $\mathcal{E}$ is as in Theorem 1.1.

Theorem 1.1 and Theorem 1.2 can also be made algorithmic, using the maximum-entropy optimization framework of [SV14].

Theorem 1.3 (see Section 5). *There is a deterministic algorithm (based on the ellipsoid algorithm) which, given an $m \times n$ totally unimodular matrix M , a vector $\mathbf{b} \in \mathbb{Z}^m$ such that $\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n \mid M\mathbf{y} = \mathbf{b}\}$ is nonempty, and an $\epsilon \in (0, 1)$, computes $\mathcal{A}$ such that $\mathcal{A} - \epsilon \leq \log \mathcal{E} \leq \mathcal{A} + \epsilon$ in time polynomial in $m, n, \log(1 + \|\mathbf{b}\|_1)$, and $\log \frac{1}{\epsilon}$. Here $\mathcal{E}$ is defined as in Theorem 1.1.*

The proofs of Theorem 1.1 and Theorem 1.2 are then based on our answer to question 2: how does the number of lattice points of a TU polytope vary with $\mathbf{b}$? The following result answers this by generalizing a question of Barvinok [Bar07] on the log-concavity of the counts of contingency tables with various marginals. Our result in fact holds for **unimodular polytopes**, which are defined like totally unimodular polytopes except that the defining matrix M is only required to be unimodular.

Theorem 1.4 (Theorem 3.1). *Let M be an $m \times n$ (not necessarily totally) unimodular matrix. For all $\mathbf{b}, \mathbf{b}^-, \mathbf{b}^+ \in \mathbb{Z}^m$ such that $\mathbf{b} = \frac{\mathbf{b}^- + \mathbf{b}^+}{2}$, we have*

$$\#\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n \mid M\mathbf{y} = \mathbf{b}\}^2 \geq \#\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n \mid M\mathbf{y} = \mathbf{b}^-\} \cdot \#\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n \mid M\mathbf{y} = \mathbf{b}^+\},$$

given that all of these fibers are finite.

We note that Theorem 1.4 immediately implies a recent conjecture for TU polytopes on the log-concavity of Ehrhart evaluations. This conjecture comes from [FH24], where the authors in fact make the conjecture for all IDP lattice polytopes (though this is now false in general by [Fer26]). Recall that, given a lattice polytope P , the Ehrhart polynomial $E_P(t)$ is the unique polynomial such that $E_P(k) = \#(kP \cap \mathbb{Z}^n)$ for all $k \in \mathbb{Z}_{\geq 0}$.

Corollary 1.5 (Conjecture 1.2 of [FH24]). *For all (not necessarily totally) unimodular polytopes P , we have $E_P(k)^2 \geq E_P(k-1) \cdot E_P(k+1)$ for all $k \in \mathbb{Z}_{\geq 1}$.*

Beyond our main results, we can also prove slightly stronger results under further restrictions. We give various bounds in Section 4, and we give one such result here to demonstrate the sort of thing we can achieve.

Theorem 1.6 (Theorem 4.14). *Let M be an $m \times n$ totally unimodular matrix with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ such that $\mathbf{a}_i \neq \mathbf{0}$ for all $i \in [n]$, and let $\mathbf{s}_i$ be the sum of the first i rows of M for $i \in [m]$. If $\mathbf{s}_i \geq \mathbf{0}$ for all $i \in [m]$ (for example, if M has only 0/1 entries) then for all $\mathbf{b} \in \mathbb{Z}^m$ we have*

$$\mathcal{E} \geq \#\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n \mid M\mathbf{y} = \mathbf{b}\} \geq \frac{|\sigma_m|^{\|\sigma_m\|}}{(1 + |\sigma_m|)^{1 + \|\sigma_m\|}} \left[\prod_{i=1}^{m-1} \frac{|\sigma_i|^{\|\sigma_i\|}}{(1 + |\sigma_i|)^{1 + \|\sigma_i\|}} \right]^2 \mathcal{E},$$

where $\sigma_i = b_1 + \dots + b_i$ for all $i \in [m]$ and

$$\mathcal{E} = \inf_{\substack{\mathbf{x} > 0 \\ \forall j \mathbf{x}^{\mathbf{a}_j} < 1}} \mathbf{x}^{-\mathbf{b}} \prod_{j=1}^n \frac{1}{1 - \mathbf{x}^{\mathbf{a}_j}} = \inf_{\mathbf{x} > 0} \mathbf{x}^{-\mathbf{b}} \prod_{j=1}^n \sum_{k=0}^{\infty} \mathbf{x}^{k\mathbf{a}_j}$$

is a convex program up to log-log transformation.

1.2 An Open Question

Theorem 1.4 is in fact based upon a stronger coefficient-wise positivity result (Theorem 3.1) which follows from the bijective proof which we give in Section 3. This theorem raises an open question, which is the natural generalization of Barvinok's question in [Bar07].

Question 1.7. *Let $M \in \mathbb{Z}^{m \times n}$ be a unimodular matrix and μ a probability distribution on $\mathbb{Z}^m$ (perhaps with finite support) such that its mean is $\mathbf{b} \in \mathbb{Z}^m$. Is it that case that*

$$\#\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n \mid M\mathbf{y} = \mathbf{b}\} \geq \prod_{\mathbf{a} \in \mathbb{Z}^m} \#\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n \mid M\mathbf{y} = \mathbf{a}\}^{\mu(\mathbf{a})}?$$

Note that for functions/distributions over $\mathbb{R}^m$, Theorem 1.7 would immediately follow from Theorem 1.4. However, for functions/distributions over $\mathbb{Z}^m$, convexity on lines does not imply global convexity in general.

We believe that a positive answer to Theorem 1.7 would enable us to obtain stronger capacity bounds (hopefully simply exponential) in general, using the following idea. Currently, most capacity bounds are proved inductively on each variable of a given polynomial or generating series. Because of this, the bounds often depend on distances of the desired marginals or degree vector (e.g. the $\mathbf{b}$ vector) to the boundary of the Newton polyhedron in the coordinate directions. The coordinate direction distance can be very bad for general TU polytopes, and so we need to adapt the application of the capacity bounds to handle this.

Given the inequalities of Theorem 1.7, it is perhaps possible that this capacity induction argument may be able to be exchanged with a geometric argument related to distances to the boundary of the Newton polyhedron in any choice of directions. This would yield much better bounds and approximation factors, if possible.

1.3 Previous and Related Work

Bounding and approximately counting lattice points in totally unimodular polytopes has attracted considerable attention. A variety of bounds and approximation algorithms, both randomized and deterministic, are known for important classes of such polytopes. On the randomized side, fully polynomial randomized approximation schemes (FPRASs) have been obtained for bipartite perfect matchings [JSV04], binary contingency tables [BBV07], and nonnegative integer contingency tables under restrictions such as a fixed number of rows or sufficiently large margins [CD03, DKM97, Mor02]. Related results apply to cell-bounded contingency tables with a fixed number of rows and to integral flows with sufficiently large tight capacities [CDR10]. For certain classes of smooth margins, a randomized approximation algorithm runs in time quasipolynomial in the total sum of the table entries [BLSY10]. More recently, rapid mixing of the natural swap chain was established for binary contingency tables with arbitrary feasible margins [FQW26].

On the deterministic side, matrix scaling and polynomial capacity methods give polynomial-time exponential approximations for counting bipartite perfect matchings [LSW00, Gur06, Gur08], with subsequent improvements [GS14, AR25]. Recent preprints obtain a subexponential approximation factor [KLM26] and a $(1 + \varepsilon)^n$ approximation factor for every fixed $\varepsilon > 0$ [DJ26], where n is the number of vertices on each side of the bipartition. Convex optimization and polynomial capacity methods also give (often simply) exponential estimates for contingency tables [Bar09, Bar10, Gur15, BLP23] and integer flows [Bar09, LM26, LMY26]. Computationally efficient asymptotic formulas are available in suitable smooth-margin regimes [BH12]. For contingency tables with a fixed number of rows, deterministic fully polynomial-time approximation schemes (FPTASs) are known [GKM⁺11], with improved running times in the two-row case [AH21].

Beyond these special classes, other positive results include polynomial-time exact counting in fixed dimension [Bar94], exact counting methods for nonnegative totally unimodular systems with a fixed number of equations [Mou00], algebraic methods for computing counting polynomials for unimodular systems [DS03], and maximum-entropy Gaussian approximations under well-roundedness assumptions [BH10].

In contrast, no fully polynomial randomized or deterministic approximation scheme is known for arbitrary totally unimodular polytopes; such an algorithm would in particular yield an FPRAS for counting independent sets in bipartite graphs [GJ23]. This motivates seeking weaker approximation guarantees that apply uniformly to all bounded TU polytopes, which is what we prove here.

1.4 AI Declaration

The AI tool we specifically used was ChatGPT 6 Astra. The authors have verified and take full responsibility for everything written in the paper.

Before the first draft. The resolution of Barvinok’s conjecture/question on the log-concavity on lines of the contingency tables counts (Theorem 1.4) was the main AI input to this paper. We generalized this result to all TU polytopes with upper and lower entrywise bounds on the counted lattice points (Theorem 3.1), and in fact the AI suggested in a remark that this should be possible. Beyond that, the literature review and much of the associated parts of the introduction were written with the help of AI.

After the first draft. We used AI to check for errors or to make suggestions on the flow or style of the paper. From this we fixed a number of errors which were mostly minor; the main issue that was fixed here were proofs regarding limiting arguments for capacity and for sequences of Laurent polynomials and series. In particular Theorem 4.11 and the surrounding results were added at this point. Additionally, AI also suggested a way to improve the bound in Theorem 1.1 via Theorem 4.3. We also originally had referenced [SV14, SV19] as the way to prove algorithmic efficiency in Theorem 1.3, but we had left out the details. We used AI to fill in those details, and now we have the proof of Theorem 1.3 in Section 5. Further, AI also found that the central-fiber argument in our (AI-assisted) proof of Theorem 1.4 has a direct antecedent in the graph-orientation argument of [CI20]. Finally, AI also suggested many changes to the structure, ordering, and content of the introduction and title. We took many of those suggestions, and then wrote the new introduction ourselves.

2 VLC Laurent polynomials

Gurvits’ capacity function has been widely used to obtain lower bounds for various interesting combinatorial quantities. The underlying idea of these results is that if a (Laurent) polynomial has a certain log-concavity property, then one could use the theory of log-concave polynomials to derive lower bounds for its coefficients. However, currently studied log-concavity properties (e.g., Lorentzian, denormalized/dually Lorentzian, etc.) are all conditions which are too strong to apply to TU polytopes in general.

Thus, in Section 2.1 we introduce a class of Laurent polynomials with weaker log-concavity properties, for which capacity lower bounds are still applicable. We further explore basic properties of this class throughout Section 2.2. We then turn our focus to unimodular and totally unimodular (TU) matrices in Section 3 and prove that the cardinalities of the fibers of a unimodular polytope are log-concave along lines.

2.1 Definition and capacity bounds

There are two key steps in proving capacity bounds for a denormalized Lorentzian polynomial p : first that any evaluation of p in all but one variable is a univariate polynomial with log-concave coefficients, and second that $[x_i^{\alpha_i}]p$ satisfies the first condition for any i, α_i . We will take these two properties and define a new class of Laurent polynomials. Capacity lower bounds then naturally extend to this class.

Recall that a sequence $\{a_k\}_{k \in \mathbb{Z}}$ of non-negative numbers is said to be log-concave if it has no internal zeros and $a_k^2 \geq a_{k-1} \cdot a_{k+1}$ for any valid index k . A Laurent polynomial is a sum $p(\mathbf{x}) = \sum_{\alpha \in \mathbb{Z}^n} p_\alpha \mathbf{x}^\alpha$ with a finite support, where the support of p denoted by $\text{supp}(p)$ is the set $\{\alpha \in \mathbb{Z}^n : p_\alpha \neq 0\}$. We further say that a univariate Laurent polynomial $p(x) = \sum_{k \in \mathbb{Z}} p_k x^k$ has a log-concave coefficient sequence if $\{p_k\}_{k \in \mathbb{Z}}$ is a log-concave sequence.

Definition 2.1. Define:

$$\text{VLC}_1 = \{p(x) \in \mathbb{R}[x^{\pm 1}] : p \text{ has a log-concave coefficient sequence}\}.$$

and for any $n \geq 2$, let VLC_n contain all n -variate Laurent polynomials $p(x_1, \dots, x_n) = \sum_{\alpha \in \mathbb{Z}^n} p_\alpha \mathbf{x}^\alpha$ with non-negative coefficients such that:

- Evaluating p at any $n - 1$ variables with positive real numbers belongs to VLC_1 ,
- $[x_i^{\alpha_i}]p$ is in VLC_{n-1} for any $i \in [n]$ and $\alpha_i \in \mathbb{Z}$.

Equivalently, $p(x_1, \dots, x_n) \in \mathbb{R}_{\geq 0}[\mathbf{x}, \mathbf{x}^{-1}]$ is an element of VLC_n if for any $S \subseteq [n]$ and any $\alpha \in \mathbb{Z}^S$, any positive evaluation of $[\prod_{i \in S} x_i^{\alpha_i}]p$ in all but one variable has log-concave coefficients. In a sense, the coefficients of p are log-concave after applying certain variable-wise operations, hence the name VLC.

More generally, we will say that a polynomial $p(x_1, \dots, x_n)$ is VLC if it belongs to VLC_n .

Lorentzian polynomials and denormalized Lorentzian polynomials are both subclasses of VLC (see [BH20, §3.1]). In fact, this class of Laurent polynomials is crafted in a way that the capacity bound for the coefficients of denormalized Lorentzian Polynomials extends naturally to VLC Laurent polynomials.

Proposition 2.2. Suppose $p(x_1, \dots, x_n) \in \text{VLC}_n$ and $\alpha \in \mathbb{Z}^n$, then:

$$[\mathbf{x}^\alpha]p(\mathbf{x}) \geq \text{cap}_\alpha(p) \cdot \prod_{i=1}^n \frac{|\beta_i - \alpha_i|^{\beta_i - \alpha_i}}{(1 + |\beta_i - \alpha_i|)^{1 + |\beta_i - \alpha_i|}}$$

where β_i is either a lower bound or an upper bound for $\text{Deg}_i^\alpha(p)$, the set of degrees of x_i appearing in $[x_{i+1}^{\alpha_{i+1}} \dots x_n^{\alpha_n}]p$.

Proof. The proof is done by induction over the number of variables of p . For $n = 1$, the proof follows from the case for denormalized Lorentzian Laurent

polynomials by [LMY26, Lemma 4.20] (see also [BLP23, Lemma 5.7]). Note that the homogenization of the polynomial $\sum_{k \in \mathbb{Z}} a_k x^k$ is denormalized Lorentzian iff $\{a_k\}_{k \in \mathbb{Z}}$ is a log-concave sequence, i.e. $\sum_{k \in \mathbb{Z}} a_k x^k$ is VLC.

Suppose $n \geq 2$. Then for any $(r_1, \dots, r_{n-1}) \in \mathbb{R}_{>0}^{n-1}$:

$$\begin{aligned} \text{cap}_\alpha(p) &\leq \inf_{x_n > 0} \frac{p(r_1, \dots, r_{n-1}, x_n)}{r_1^{\alpha_1} \dots r_{n-1}^{\alpha_{n-1}} x_n^{\alpha_n}} \\ &= \frac{1}{r_1^{\alpha_1} \dots r_{n-1}^{\alpha_{n-1}}} \inf_{x_n > 0} \frac{p(r_1, \dots, r_{n-1}, x_n)}{x_n^{\alpha_n}} \\ &= \frac{1}{r_1^{\alpha_1} \dots r_{n-1}^{\alpha_{n-1}}} \text{cap}_{\alpha_n} p(r_1, \dots, r_{n-1}, x_n) \\ &\leq \frac{1}{r_1^{\alpha_1} \dots r_{n-1}^{\alpha_{n-1}}} \cdot \frac{(1 + |\beta_n - \alpha_n|)^{1+|\beta_n - \alpha_n|}}{|\beta_n - \alpha_n|^{|\beta_n - \alpha_n|}} \cdot [x_n^{\alpha_n}] p(r_1, \dots, r_{n-1}, x_n) \end{aligned}$$

Since this is true for any $(r_1, \dots, r_{n-1})$, we conclude:

$$\begin{aligned} \text{cap}_\alpha(p) &\leq \frac{(1 + |\beta_n - \alpha_n|)^{1+|\beta_n - \alpha_n|}}{|\beta_n - \alpha_n|^{|\beta_n - \alpha_n|}} \cdot \inf_{r_1, \dots, r_{n-1} > 0} \frac{[x_n^{\alpha_n}] p(r_1, \dots, r_{n-1}, x_n)}{r_1^{\alpha_1} \dots r_{n-1}^{\alpha_{n-1}}} \\ &= \frac{(1 + |\beta_n - \alpha_n|)^{1+|\beta_n - \alpha_n|}}{|\beta_n - \alpha_n|^{|\beta_n - \alpha_n|}} \cdot \text{cap}_{(\alpha_1, \dots, \alpha_{n-1})}([x_n^{\alpha_n}] p). \end{aligned}$$

The proof is then completed by the induction hypothesis for $[x_n^{\alpha_n}] p$. $\square$

2.2 Examples and properties

As mentioned before, any denormalized Lorentzian polynomial is VLC. It is easy to see that any “shift” of a denormalized Lorentzian polynomial is also VLC, i.e. $\mathbf{x}^\alpha p(\mathbf{x})$ is VLC if p is denormalized Lorentzian. However, the class of VLC Laurent polynomials contains a much larger set of Laurent polynomials. The following is an example of a Laurent polynomial that is not necessarily denormalized Lorentzian but is VLC:

Example 2.3. Let $n, k \in \mathbb{Z}_{\geq 0}$ and $\mathbf{b} \in \{0, \pm 1\}^n$, then:

$$p(x_1, \dots, x_n) = \sum_{t=0}^k \mathbf{x}^{t \cdot \mathbf{b}}$$

is VLC. We proceed by induction on n . If $n = 1$, the statement is trivial since the coefficients sequence of p is the all ones sequence. Now suppose $n \geq 2$ and assume the statement holds for $n - 1$. Let us check both of the conditions of Theorem 2.1 for p .

First evaluate p at all but one variable, say x_i . It is easy to check that the coefficients of the polynomial $p(\mathbf{x})|_{x_j=r_j \forall j \neq i}$ form a geometric sequence and thus are log-concave. Furthermore, if $b_i \neq 0$, the Laurent polynomial $[x_i^{\alpha_i}] p(\mathbf{x})$ is either the zero polynomial or just a monomial and is VLC. If $b_i = 0$ however,

$[x_i^{\alpha_i}]p$ is really a Laurent polynomial in $n - 1$ variables which is VLC by the induction hypothesis.

As another example, suppose $\mathbf{b}_1, \mathbf{b}_2 \in \{0, 1\}^n$, then the series:

$$q(\mathbf{x}) = \sum_{t=0}^k \mathbf{x}^{t \cdot \mathbf{b}_1} \cdot \sum_{t=0}^k \mathbf{x}^{t \cdot \mathbf{b}_2}$$

can also be shown to be VLC by basic algebra and some extra effort. We have refrained from including the details of this proof for brevity. One could see that this polynomial is indeed VLC by Theorem 3.8.

For general $\mathbf{b}, \mathbf{b}_1, \mathbf{b}_2$, the supports of the Laurent polynomials p and q in Theorem 2.3 need not be M -convex. Thus, neither is necessarily a denormalized Lorentzian Laurent polynomial. This raises the question of characterizing the support of an VLC Laurent polynomial.

Proposition 2.4. *If $p(\mathbf{x}) = \sum_{\alpha \in \mathbb{Z}^n} p_{\alpha} \mathbf{x}^{\alpha}$ is an VLC Laurent polynomial, then its support is a finite saturated set.*

$S \subseteq \mathbb{Z}^n$ is called a saturated set if $S = \text{conv}(S) \cap \mathbb{Z}^n$, where $\text{conv}(S)$ is the convex hull of S .

Before proving this proposition, let us recall a well-known fact about $\text{newt}(p) := \text{conv}(\text{supp}(p))$, also known as the Newton polytope of p .

Fact 2.5 ([AO17, Fact 2.18], see also [LMY26, Lemma 4.15]). *If p is a nonzero Laurent polynomial (or a Laurent series with a nonempty domain of convergence and a closed Newton polyhedron) with nonnegative coefficients, then $\text{cap}_{\alpha}(p) > 0$ iff $\alpha \in \text{newt}(p)$.*

Proof of Theorem 2.4. It suffices to show that $p_{\alpha} > 0$ for any $\alpha \in \text{newt}(p) \cap \mathbb{Z}^n$. By Theorem 2.5, for any such α , $\text{cap}_{\alpha}(p) > 0$. Thus, by Theorem 2.2, we have:

$$p_{\alpha} \geq \text{cap}_{\alpha}(p) \cdot \prod_{i=1}^n \frac{|\beta_i - \alpha_i|^{\beta_i - \alpha_i}}{(1 + |\beta_i - \alpha_i|)^{1 + |\beta_i - \alpha_i|}} > 0.$$

□

One would hope that any finite saturated set would be the support of some VLC Laurent polynomial, but this is not the case. For instance, the set $S = \{(0, 0), (1, 0), (2, 1)\}$ is a saturated set, but the polynomial $p(x, y) = a + bx + cx^2y$ is not VLC no matter the choice of $a, b, c > 0$. This is because evaluating y at a large enough $r > 0$ breaks log-concavity of the sequence a, b, cr . We currently do not have a necessary and sufficient condition for a set S to arise the support of an VLC Laurent polynomial.

Proposition 2.6. *The class of VLC Laurent polynomials is closed under scaling variables.*

Proof. We proceed by induction on the number of variables. If $p(x) = \sum_{k \in \mathbb{Z}} p_k x^k$ has a log-concave coefficient sequence, then so does $p(cx) = \sum_{k \in \mathbb{Z}} c^k p_k x^k$, and the induction basis holds.

Let $n \geq 2$ and let $p(x_1, \dots, x_n)$ be an VLC Laurent polynomial and let $c > 0$. We want to show that $q(\mathbf{x}) = p(cx_1, x_2, \dots, x_n)$ is also an VLC Laurent polynomial. Any evaluation of q at all but the variable x_i where $i \neq 1$ has log-concave coefficients, since:

$$q(\mathbf{x})|_{x_j=r_j, j \neq i} = p(\mathbf{x})|_{x_1=cr_1, x_j=r_j j \neq i, 1}.$$

If q is evaluated at all but variables except x_1 , then:

$$q(x_1, r_2, \dots, r_n) = p(cx_1, r_2, \dots, r_n) = \sum_{k \in \mathbb{Z}} p_k c^k x_1^k,$$

where p_k is the coefficient of x_1^k in $p(x_1, r_2, \dots, r_n)$. Since p is VLC, the sequence $\{p_k\}_{k \in \mathbb{Z}}$ is log-concave, and therefore, so is the sequence $\{c^k p_k\}_{k \in \mathbb{Z}}$. Hence any evaluation of q in $n - 1$ variables has log-concave coefficients.

Furthermore, we know that $[x_i^\alpha]p$ is an VLC Laurent polynomial in $n - 1$ variables for any $\alpha \in \mathbb{Z}$ and $i \in [n]$. By the induction hypothesis, for any $i \neq 1$, the Laurent polynomial $[x_i^\alpha]q = ([x_i^\alpha]p)(cx_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$ is VLC as well. Lastly, $[x_1^\alpha]q = c^\alpha [x_1^\alpha]p$ is VLC too and we are done. $\square$

Proposition 2.7. *If a sequence $\{p_i(\mathbf{x})\}_{i \geq 0}$ of VLC_n Laurent polynomials converges coefficient-wise to a Laurent polynomial $p(\mathbf{x})$ such that $\text{supp}(p_i) \subseteq \text{supp}(p)$ for all $i \geq 0$, then $p \in \text{VLC}_n$.*

Proof. Proceed with induction on n . The statement for $n = 1$ follows from the fact that the limit of any log-concave sequence is a log-concave sequence. Let $n \geq 2$. Suppose $r_2, \dots, r_n > 0$. Then:

$$p(x_1, r_2, \dots, r_n) = \lim_{i \rightarrow \infty} p_i(x_1, r_2, \dots, r_n),$$

since each $p_i(x_1, r_2, \dots, r_n)$ has a log-concave coefficient sequence, the coefficient sequence of $p(x_1, r_2, \dots, r_n)$ is log-concave as well. Moreover, for any $\alpha \in \mathbb{Z}$, $j \in [n]$:

$$[x_j^\alpha]p = \lim_{i \rightarrow \infty} [x_j^\alpha]p_i.$$

Since $\text{supp}([x_j^\alpha]p_i) \subseteq \text{supp}([x_j^\alpha]p)$ for all $i \geq 0$, $[x_j^\alpha]p$ is VLC_{n-1} by the induction hypothesis, finishing the proof. $\square$

3 Unimodular and Totally unimodular (TU) polytopes

The goal of this section is to prove the following log-concavity statement:

$$\left(\sum_{\substack{M\mathbf{y} = \frac{\mathbf{b}^+ + \mathbf{b}^-}{2} \\ \mathbf{k}_1 \leq \mathbf{y} \leq \mathbf{k}_2}} \mathbf{w}^{\mathbf{y}} \right)^2 \geq \sum_{\substack{M\mathbf{y} = \mathbf{b}^+ \\ \mathbf{k}_1 \leq \mathbf{y} \leq \mathbf{k}_2}} \mathbf{w}^{\mathbf{y}} \cdot \sum_{\substack{M\mathbf{y} = \mathbf{b}^- \\ \mathbf{k}_1 \leq \mathbf{y} \leq \mathbf{k}_2}} \mathbf{w}^{\mathbf{y}},$$

where M is an $m \times n$ unimodular matrix and $\mathbf{b}^+, \mathbf{b}^-, \frac{\mathbf{b}^+ + \mathbf{b}^-}{2} \in \mathbb{Z}^m, \mathbf{w} \in \mathbb{R}_{>0}^n$ are given vectors. Recall that a matrix M is said to be unimodular if all of its non-singular $\text{rank}(M) \times \text{rank}(M)$ submatrices have determinant ± 1 . We will provide an AI-assisted proof that this sequence is log-concave for any M and $\mathbf{b}$ in Section 3.1. In Section 3.2, we will use this log-concavity statement to show that $\sum_{0 \leq \mathbf{y} \leq \mathbf{k}} \mathbf{x}^{M\mathbf{y}}$ is an VLC Laurent polynomial for any TU matrix M . We will then derive capacity bounds for $\#\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n : M\mathbf{y} = \mathbf{b}, \mathbf{y} \leq \mathbf{k}\}$ using this Laurent polynomial in Section 4.

3.1 Mid point log-concavity for unimodular polytopes

The ideas and proofs in this section follow an AI assisted argument for midpoint log-concavity of contingency tables with tweaks to prove what we require for our purposes. In particular, the proofs of Theorem 3.1, Theorem 3.2 and Theorem 3.3 are AI-assisted. We note that the pair-decomposition and induction used in Theorem 3.2 and Theorem 3.3 generalize the argument from Section 5 of [CI20] from graph orientations to lattice points in boxes subject to unimodular linear constraints.

For any unimodular matrix $M \in \mathbb{Z}^{m \times n}$ and any vectors $\mathbf{b}, \mathbf{a}, \mathbf{k}_1, \mathbf{k}_2$ of proper dimensions, Define:

$$\begin{aligned} \mathcal{B}_{(\mathbf{k}_1, \mathbf{k}_2)}(M, \mathbf{b}) &= \{\mathbf{y} : M\mathbf{y} = \mathbf{b}, \mathbf{k}_1 \leq \mathbf{y} \leq \mathbf{k}_2\}, \\ B_{(\mathbf{k}_1, \mathbf{k}_2)}(M, \mathbf{b}) &= |\mathcal{B}_{(\mathbf{k}_1, \mathbf{k}_2)}(M, \mathbf{b}) \cap \mathbb{Z}^n|, \\ \mathcal{E}(M, \mathbf{a}) &= \{\mathbf{y} : M\mathbf{y} = \frac{M\mathbf{a}}{2}, y_i \in \{0, a_i\} \forall i \in [n]\}, \\ E(M, \mathbf{a}) &= |\mathcal{E}(M, \mathbf{a})|, \\ T(M, \mathbf{b}, \mathbf{k}_1, \mathbf{k}_2; \mathbf{w}) &= \sum_{\substack{\mathbf{k}_1 \leq \mathbf{y} \leq \mathbf{k}_2 \\ M\mathbf{y} = \mathbf{b}}} \mathbf{w}^{\mathbf{y}}. \end{aligned}$$

When the choice of the matrix M is clear from context, we will omit M from the notation in all of the definitions above. Note that if $E(M, \mathbf{a}) \neq 0$, then we must have $M\mathbf{a} \in (2\mathbb{Z})^m$. We aim to prove the following theorem:

Theorem 3.1. Suppose $\mathbf{b}, \mathbf{b}^+, \mathbf{b}^-$ are integer vectors such that $\mathbf{b} = \frac{\mathbf{b}^+ + \mathbf{b}^-}{2}$. Then for any $\mathbf{k}_1, \mathbf{k}_2$:

$$T(\mathbf{b}, \mathbf{k}_1, \mathbf{k}_2; \mathbf{w})^2 - T(\mathbf{b}^+, \mathbf{k}_1, \mathbf{k}_2; \mathbf{w}) \cdot T(\mathbf{b}^-, \mathbf{k}_1, \mathbf{k}_2; \mathbf{w})$$

is a Laurent polynomial in $\mathbf{w}$ with nonnegative coefficients.

In fact, we will explicitly express the coefficients of $T(\mathbf{b}, \mathbf{k}_1, \mathbf{k}_2; \mathbf{w})^2 - T(\mathbf{b}^+, \mathbf{k}_1, \mathbf{k}_2; \mathbf{w}) \cdot T(\mathbf{b}^-, \mathbf{k}_1, \mathbf{k}_2; \mathbf{w})$ in terms of $B_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b})$, $B_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b}^+)$ and $B_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b}^-)$ and prove that these coefficients are non-negative using Theorem 3.2, Theorem 3.3 and Theorem 3.4. We then use Theorem 3.1 to prove that the Ehrhart of a unimodular polytope has log-concave evaluations, see Theorem 3.5. We explain why our methods do not work for a general IDP polytope in Theorem 3.6.

Lemma 3.2. For any integer vector $\mathbf{b} \in \mathbb{Z}^m$, we have:

$$B_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b})^2 = \sum_{\substack{\mathbf{a} \in \mathbb{Z}^n \\ \mathbf{0} \leq \mathbf{a} \leq \mathbf{k}_2 - \mathbf{k}_1, \\ E(\mathbf{a}) > 0}} E(\mathbf{a}) B_{(\mathbf{k}_1, \mathbf{k}_2 - \mathbf{a})} \left(\mathbf{b} - \frac{M\mathbf{a}}{2} \right).$$

Proof. It suffices to find a bijection:

$$[\mathcal{B}_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b}) \times \mathcal{B}_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b})] \cap \mathbb{Z}^{2n} \xrightarrow{\varphi} \left\{ (\mathbf{a}, \mathbf{p}, \mathbf{m}) : \begin{array}{l} \mathbf{0} \leq \mathbf{a} \leq \mathbf{k}_2 - \mathbf{k}_1, \mathbf{a} \in \mathbb{Z}^n \\ \mathbf{p} \in \mathcal{E}(\mathbf{a}) \\ \mathbf{m} \in \mathcal{B}_{(\mathbf{k}_1, \mathbf{k}_2 - \mathbf{a})}(\mathbf{b} - \frac{M\mathbf{a}}{2}) \cap \mathbb{Z}^n \end{array} \right\}.$$

For any $\mathbf{x}, \mathbf{y} \in \mathcal{B}_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b}) \cap \mathbb{Z}^n$, define:

$$\mathbf{a} = |\mathbf{x} - \mathbf{y}|, \quad \mathbf{p} = (\mathbf{x} - \mathbf{y})_+, \quad \mathbf{m} = \min(\mathbf{x}, \mathbf{y})$$

where for any number $r \in \mathbb{R}$, we define $(r)_+ = 0$ if $r < 0$ and $(r)_+ = r$ otherwise. The following facts are easy to check:

1. $\mathbf{x} = \mathbf{m} + \mathbf{p}$,
2. $\mathbf{y} = \mathbf{m} + \mathbf{a} - \mathbf{p}$,
3. $2\mathbf{p} = \mathbf{a} + \mathbf{x} - \mathbf{y}$ so $M\mathbf{p} = \frac{M\mathbf{a}}{2}$ and further $p_i \in \{0, a_i\}$, thus $\mathbf{p} \in \mathcal{E}(\mathbf{a})$,
4. $2\mathbf{m} = \mathbf{x} + \mathbf{y} - \mathbf{a}$, so $M\mathbf{m} = \mathbf{b} - \frac{M\mathbf{a}}{2}$. Moreover $\mathbf{k}_1 \leq \mathbf{m} \leq \mathbf{k}_2 - \mathbf{a}$. Thus $\mathbf{m} \in \mathcal{B}_{(\mathbf{k}_1, \mathbf{k}_2 - \mathbf{a})}(\mathbf{b} - \frac{M\mathbf{a}}{2}) \cap \mathbb{Z}^n$.

Let: $\varphi(\mathbf{x}, \mathbf{y}) = (\mathbf{a}, \mathbf{p}, \mathbf{m})$. Then φ is a bijection between the desired sets by 1 through 4. $\square$

Lemma 3.3. Given integer vectors $\mathbf{k}_1 \leq \mathbf{k}_2$ and $\mathbf{b}$ with proper dimensions, define $\mathbf{k} := \mathbf{k}_2 + \mathbf{k}_1$. If $M\mathbf{k}$ is even, then we have:

$$B_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b}) \leq B_{(\mathbf{k}_1, \mathbf{k}_2)} \left(\frac{M\mathbf{k}}{2} \right).$$

Proof. Proof is done by induction over $|\mathbf{k}_2 - \mathbf{k}_1|$. Suppose $|\mathbf{k}_2 - \mathbf{k}_1| = 0$, i.e. $\mathbf{k}_2 = \mathbf{k}_1$. Then $B_{(\mathbf{k}_1, \mathbf{k}_2)}(\frac{M\mathbf{k}}{2}) = 1$, and $B_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b})$ is either zero or one, depending on whether $M\mathbf{k}_1 = \mathbf{b}$ or not. Thus the base case holds.

By Theorem 3.2, write:

$$B_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b})^2 = \sum_{\substack{\mathbf{a} \in \mathbb{Z}^n \\ \mathbf{0} \leq \mathbf{a} \leq \mathbf{k}_2 - \mathbf{k}_1 \\ E(\mathbf{a}) > 0}} E(\mathbf{a}) B_{(\mathbf{k}_1, \mathbf{k}_2 - \mathbf{a})} \left(\mathbf{b} - \frac{M\mathbf{a}}{2} \right),$$

and:

$$B_{(\mathbf{k}_1, \mathbf{k}_2)} \left(\frac{M\mathbf{k}}{2} \right)^2 = \sum_{\substack{\mathbf{a} \in \mathbb{Z}^n \\ \mathbf{0} \leq \mathbf{a} \leq \mathbf{k}_2 - \mathbf{k}_1 \\ E(\mathbf{a}) > 0}} E(\mathbf{a}) B_{(\mathbf{k}_1, \mathbf{k}_2 - \mathbf{a})} \left(\frac{M(\mathbf{k} - \mathbf{a})}{2} \right).$$

The term $\mathbf{a} = \mathbf{0}$ contributes $B_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b})$ and $B_{(\mathbf{k}_1, \mathbf{k}_2)}(\frac{M\mathbf{k}}{2})$ to the right hand sides of the first and the second identity respectively. By moving these terms to the left hand side, we get:

$$B_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b})^2 - B_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b}) = \sum_{\substack{\mathbf{a} \in \mathbb{Z}^n, \mathbf{a} \neq \mathbf{0} \\ \mathbf{0} \leq \mathbf{a} \leq \mathbf{k}_2 - \mathbf{k}_1 \\ E(\mathbf{a}) > 0}} E(\mathbf{a}) B_{(\mathbf{k}_1, \mathbf{k}_2 - \mathbf{a})} \left(\mathbf{b} - \frac{M\mathbf{a}}{2} \right),$$

and:

$$B_{(\mathbf{k}_1, \mathbf{k}_2)} \left(\frac{M\mathbf{k}}{2} \right)^2 - B_{(\mathbf{k}_1, \mathbf{k}_2)} \left(\frac{M\mathbf{k}}{2} \right) = \sum_{\substack{\mathbf{a} \in \mathbb{Z}^n, \mathbf{a} \neq \mathbf{0} \\ \mathbf{0} \leq \mathbf{a} \leq \mathbf{k}_2 - \mathbf{k}_1 \\ E(\mathbf{a}) > 0}} E(\mathbf{a}) B_{(\mathbf{k}_1, \mathbf{k}_2 - \mathbf{a})} \left(\frac{M(\mathbf{k} - \mathbf{a})}{2} \right).$$

By induction hypothesis, for any $\mathbf{0} \leq \mathbf{a} \leq \mathbf{k}_2 - \mathbf{k}_1$, $\mathbf{a} \neq \mathbf{0}$:

$$B_{(\mathbf{k}_1, \mathbf{k}_2 - \mathbf{a})} \left(\mathbf{b} - \frac{M\mathbf{a}}{2} \right) \leq B_{(\mathbf{k}_1, \mathbf{k}_2 - \mathbf{a})} \left(\frac{M(\mathbf{k} - \mathbf{a})}{2} \right),$$

implying that:

$$B_{(\mathbf{k}_1, \mathbf{k}_2)} \left(\frac{M\mathbf{k}}{2} \right)^2 - B_{(\mathbf{k}_1, \mathbf{k}_2)} \left(\frac{M\mathbf{k}}{2} \right) \geq B_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b})^2 - B_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b}).$$

This conclusion completes the proof except for the case that $B_{(\mathbf{k}_1, \mathbf{k}_2)}(\frac{M\mathbf{k}}{2}) = 0$ and $B_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b}) = 1$. However, $\mathcal{B}_{(\mathbf{k}_1, \mathbf{k}_2)}(\frac{M\mathbf{k}}{2})$ is a nonempty polytope since $\frac{\mathbf{k}}{2} \in \mathcal{B}_{(\mathbf{k}_1, \mathbf{k}_2)}(\frac{M\mathbf{k}}{2})$, and therefore has at least one vertex which is integral by Theorem 3.4. Thus $B_{(\mathbf{k}_1, \mathbf{k}_2)}(\frac{M\mathbf{k}}{2}) \geq 1$ for any $\mathbf{k}$, fixing the issue. $\square$

Lemma 3.4. Suppose $M \in \mathbb{Z}^{m \times n}$ is a unimodular matrix and $\mathbf{b} \in \mathbb{Z}^m, \mathbf{k}_1, \mathbf{k}_2 \in \mathbb{Z}^n$. If the polytope $\mathcal{B}_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b}) = \{\mathbf{y} : M\mathbf{y} = \mathbf{b}, \mathbf{k}_1 \leq \mathbf{y} \leq \mathbf{k}_2\}$ is nonempty, then all of its vertices are integral.

Proof. Let $r = \text{rank}(M)$ and fix $\mathbf{y}^*$ to be a vertex of $\mathcal{B}_{(\mathbf{k}_1, \mathbf{k}_2)}(M, \mathbf{b})$. We say that a constraint $k_{1,i} \leq y_i$ or $y_i \leq k_{2,i}$ is active at $\mathbf{y}^*$ if it holds with equality for $\mathbf{y}^*$. The equality constraints $M\mathbf{y}^* = \mathbf{b}$ are automatically active.

Without loss of generality, assume that the first r rows of M are linearly independent, and let $M_{[r]}$ be the submatrix of M containing these r rows, and let $\mathbf{b}_{[r]}$ be the vector $\mathbf{b}$ restricted to its first r entries.

Note that the normal vectors of the active constraints of $\mathcal{B}_{\mathbf{k}_1, \mathbf{k}_2}(\mathbf{b})$ at $\mathbf{y}^*$ must span $\mathbb{R}^n$. Otherwise there will exist some $\mathbf{v} \in \mathbb{R}^n$ orthogonal to all of the active constraints. Therefore, any point $\mathbf{y}^* \pm \epsilon \mathbf{v}$ belongs to P for small enough $\epsilon \in \mathbb{R}_{\geq 0}$, contradicting the assumption that $\mathbf{y}^*$ is a vertex of M .

So in addition to the r constraints of $M_{[r]}$ that are automatically active at $\mathbf{y}^*$, there are (at least) $n - r$ constraints of $\mathbf{k}_1 \leq \mathbf{y} \leq \mathbf{k}_2$ that are active at $\mathbf{y}^*$ and are linearly independent together with the equality constraints of $M_{[r]}$. After permuting the columns of M , assume that $y_i = d_i$ for $i \in [n - r]$ where $d_i \in \{k_{1,i}, k_{2,i}\}$. For simplicity, let A be the submatrix of the first $n - r$ columns of $M_{[r]}$ and B its last r columns, so that $M_{[r]} = \begin{bmatrix} A & B \end{bmatrix}$. We know that $\mathbf{y}^*$ satisfies:

$$\begin{bmatrix} A & B \\ \text{id}_{(n-r) \times (n-r)} & 0_{(n-r) \times r} \end{bmatrix} \mathbf{y}^* = \begin{bmatrix} \mathbf{b}_{[r]} \\ \mathbf{d} \end{bmatrix}, \det \begin{bmatrix} A & B \\ \text{id}_{(n-r) \times (n-r)} & 0_{(n-r) \times r} \end{bmatrix} \neq 0,$$

In particular, we must have $\det B \neq 0$. Since B is a non-singular $r \times r$ submatrix of M , its determinant is either $+1$ or -1 and B^{-1} is an integer matrix. Thus:

$$y_j^* = [B^{-1}(\mathbf{b}_{[r]} - A\mathbf{d})]_j \in \mathbb{Z}, \quad \forall j \in \{n - r + 1, \dots, n\}$$

and $\mathbf{y}^*$ is an integer point. □

We now have all the tools to prove Theorem 3.1.

Proof of Theorem 3.1. Expand $T(\mathbf{b}, \mathbf{k}_1, \mathbf{k}_2; \mathbf{w})^2 - T(\mathbf{b}^+, \mathbf{k}_1, \mathbf{k}_2; \mathbf{w}) \cdot T(\mathbf{b}^-, \mathbf{k}_1, \mathbf{k}_2; \mathbf{w})$:

$$\begin{aligned} T(\mathbf{b}, \mathbf{k}_1, \mathbf{k}_2; \mathbf{w})^2 &= \sum_{2\mathbf{k}_1 \leq \boldsymbol{\alpha} \leq 2\mathbf{k}_2} \left(\sum_{\substack{\mathbf{k}_1 \leq \mathbf{x}, \mathbf{y} \leq \mathbf{k}_2 \\ M\mathbf{x} = M\mathbf{y} = \mathbf{b} \\ \mathbf{x} + \mathbf{y} = \boldsymbol{\alpha}}} 1 \right) \mathbf{w}^\alpha \\ &= \sum_{\substack{2\mathbf{k}_1 \leq \boldsymbol{\alpha} \leq 2\mathbf{k}_2 \\ M\boldsymbol{\alpha} = 2\mathbf{b}}} \left(\sum_{\substack{\mathbf{k}_1 \leq \mathbf{x} \leq \mathbf{k}_2 \\ \mathbf{k}_1 \leq \boldsymbol{\alpha} - \mathbf{x} \leq \mathbf{k}_2 \\ M\mathbf{x} = \mathbf{b}}} 1 \right) \mathbf{w}^\alpha \\ &= \sum_{\substack{2\mathbf{k}_1 \leq \boldsymbol{\alpha} \leq 2\mathbf{k}_2 \\ M\boldsymbol{\alpha} = 2\mathbf{b}}} B_{(\max\{\mathbf{k}_1, \boldsymbol{\alpha} - \mathbf{k}_2\}, \min\{\mathbf{k}_2, \boldsymbol{\alpha} - \mathbf{k}_1\})}(\mathbf{b}) \mathbf{w}^\alpha, \end{aligned}$$

and:

$$\begin{aligned}
T(\mathbf{b}^+, \mathbf{k}_1, \mathbf{k}_2; \mathbf{w}) \cdot T(\mathbf{b}^-, \mathbf{k}_1, \mathbf{k}_2; \mathbf{w}) &= \sum_{2\mathbf{k}_1 \leq \alpha \leq 2\mathbf{k}_2} \left(\sum_{\substack{\mathbf{k}_1 \leq \mathbf{x}, \mathbf{y} \leq \mathbf{k}_2 \\ M\mathbf{x}=\mathbf{b}^+, M\mathbf{y}=\mathbf{b}^- \\ \mathbf{x}+\mathbf{y}=\alpha}} 1 \right) \mathbf{w}^\alpha \\
&= \sum_{\substack{2\mathbf{k}_1 \leq \alpha \leq 2\mathbf{k}_2 \\ M\alpha=2\mathbf{b}}} B_{(\max\{\mathbf{k}_1, \alpha-\mathbf{k}_2\}, \min\{\mathbf{k}_2, \alpha-\mathbf{k}_1\})}(\mathbf{b}^+) \mathbf{w}^\alpha.
\end{aligned}$$

where the min, max are taken coordinate-wise. For each i , if $\alpha_i \leq k_{1,i} + k_{2,i}$ then the i -th coordinate of $\max\{\mathbf{k}_1, \alpha - \mathbf{k}_2\}$ is $k_{1,i}$ and the i -th coordinate of $\min\{\mathbf{k}_2, \alpha - \mathbf{k}_1\}$ is $\alpha_i - k_{1,i}$. If $\alpha_i \geq k_{1,i} + k_{2,i}$ then the i -th coordinate of $\max\{\mathbf{k}_1, \alpha - \mathbf{k}_2\}$ is $\alpha_i - k_{2,i}$ and the i -th coordinate of $\min\{\mathbf{k}_2, \alpha - \mathbf{k}_1\}$ is $k_{2,i}$. Thus $\max\{\mathbf{k}_1, \alpha - \mathbf{k}_2\} + \min\{\mathbf{k}_2, \alpha - \mathbf{k}_1\} = \alpha$.

it follows from Theorem 3.3 that all the coefficients of $T(\mathbf{b}, \mathbf{k}_1, \mathbf{k}_2; \mathbf{w})^2 - T(\mathbf{b}^+, \mathbf{k}_1, \mathbf{k}_2; \mathbf{w}) \cdot T(\mathbf{b}^-, \mathbf{k}_1, \mathbf{k}_2; \mathbf{w})$ are nonnegative. $\square$

Corollary 3.5. *Let $M \in \mathbb{Z}^{m \times n}$ be a unimodular matrix and $\mathbf{b} \in \mathbb{Z}^m$. Suppose $P = \{\mathbf{y} \in \mathbb{R}_{\geq 0}^n : M\mathbf{y} = \mathbf{b}\}$ is a bounded polytope. Then $E_P(t)^2 \geq E_P(t-1) \cdot E_P(t+1)$ for all $t \in \mathbb{Z}_{\geq 1}$.*

Proof. By Theorem 3.1:

$$T(t\mathbf{b}, \mathbf{0}, \mathbf{k}; \mathbf{1})^2 \geq T((t-1)\mathbf{b}, \mathbf{0}, \mathbf{k}; \mathbf{1}) \cdot T((t+1)\mathbf{b}, \mathbf{0}, \mathbf{k}; \mathbf{1})$$

As $\mathbf{k}$ tends to ∞ coordinate-wise we get:

$$\#\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n : M\mathbf{y} = t\mathbf{b}\}^2 \geq \#\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n : M\mathbf{y} = (t-1)\mathbf{b}\} \cdot \#\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n : M\mathbf{y} = (t+1)\mathbf{b}\}.$$

$\square$

Remark 3.6. We remark that if M is unimodular and $\mathbf{b} \in \mathbb{Z}^m, \mathbf{k}_1, \mathbf{k}_2 \in \mathbb{Z}^n$ are given, the polytope $\mathcal{B}_{(\mathbf{k}_1, \mathbf{k}_2)}(M, \mathbf{b})$ is indeed IDP. However, not all IDP polytopes are obtained this way. Furthermore, Theorem 3.4 is not generally correct for an arbitrary IDP polytope. In particular, with the added assumption that M has full row rank, one can show that M is unimodular iff $\mathcal{B}_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b})$ has integral vertices for any choice of $\mathbf{b}, \mathbf{k}_1, \mathbf{k}_2$. Thus, the proof of Theorem 3.1 does not extend to IDP polytopes. In fact, Theorem 3.5 fails for general IDP polytopes, see [Fer26].

3.2 Solution Laurent polynomials of TU matrices

Suppose $M \in \{0, \pm 1\}^{m \times n}$ is a totally unimodular matrix. For any $\mathbf{b} \in \mathbb{Z}^m$, we want to find a lower bound for $\#\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n : M\mathbf{y} = \mathbf{b}\}$. We encode these quantities as coefficients of a Laurent polynomial and prove that this Laurent polynomial is VLC. We will then use Theorem 2.2 to obtain lower bounds for its coefficients in Section 4.

Suppose $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{Z}^m$ are the columns of M . For any integer vector $\mathbf{k} \in \mathbb{Z}_{\geq 0}^n$, define the $\mathbf{k}$ -solution Laurent polynomial of M to be:

$$P_{M,\mathbf{k}}(x_i : i \in [m]) = \prod_{j=1}^n \left(\sum_{\ell=0}^{k_j} x^{\ell \mathbf{a}_j} \right) = \sum_{\mathbf{b} \in \mathbb{Z}^m} B_{(\mathbf{0}, \mathbf{k})}(\mathbf{b}) x^{\mathbf{b}},$$

where $B_{(\mathbf{k}_1, \mathbf{k}_2)}(\mathbf{b}) = \#\{\mathbf{y} \in \mathbb{Z}^n : M\mathbf{y} = \mathbf{b}, \mathbf{k}_1 \leq \mathbf{y} \leq \mathbf{k}_2\}$, as defined in Section 3.1.

The goal of this section is to prove that $P_{M,\mathbf{k}}$ is VLC for any TU matrix M and $\mathbf{k} \in \mathbb{Z}^n$. See Theorem 3.9 on why we cannot prove a similar statement for unimodular matrices.

Let us first derive a simpler equivalent condition for $P_{M,\mathbf{k}}$ to be VLC. For the sake of simplicity, if M is an $m \times n$ matrix and $S \subseteq [m]$, we denote by M_S the submatrix of M consisting of the rows indexed by S .

Proposition 3.7. *Suppose M is a TU matrix. Then $P_{M,\mathbf{k}}$ is VLC if and only if the following condition holds:*

Choose an arbitrary subset $S \subseteq [m]$ of the rows of M . Let $B = M_S$ and $A = M_{[m] \setminus S}$. Further choose an arbitrary $\mathbf{b} \in \mathbb{Z}^{|S|}$. Then any positive evaluation of the Laurent polynomial $\sum_{\substack{\mathbf{0} \leq \mathbf{y} \leq \mathbf{k} \\ B\mathbf{y} = \mathbf{b}}} x^{A\mathbf{y}}$ in all but one variable has log-concave coefficients.

Proof. By Theorem 2.1, $P_{M,\mathbf{k}}$ is VLC if for any $S \subseteq [m]$ and $\mathbf{b} \in \mathbb{Z}^{|S|}$, any positive evaluation of $[x_S^{\mathbf{b}}]p_{M,\mathbf{k}}$ in all but one variable has log-concave coefficients. We can write:

$$\begin{aligned} [x_S^{\mathbf{b}}]p_{M,\mathbf{k}}(\mathbf{x}) &= [x_S^{\mathbf{b}}] \sum_{\mathbf{0} \leq \mathbf{y} \leq \mathbf{k}} x^{M\mathbf{y}} \\ &= \sum_{\substack{\mathbf{0} \leq \mathbf{y} \leq \mathbf{k} \\ B\mathbf{y} = \mathbf{b}}} x^{A\mathbf{y}}, \end{aligned}$$

completing the proof. $\square$

Theorem 3.7 already reveals the connection between VLC properties of $\mathbf{k}$ -solution Laurent polynomials of TU matrices and midpoint log-concavity for TU polytopes. The following proposition completes this connection.

Proposition 3.8. *Suppose $\begin{bmatrix} A \\ B \end{bmatrix}$ is a TU matrix and assume A is an $m \times n$ matrix. Then any positive evaluation of $\sum_{\substack{\mathbf{0} \leq \mathbf{y} \leq \mathbf{k} \\ B\mathbf{y} = \mathbf{b}}} x^{A\mathbf{y}}$ in all but one variable has log-concave coefficients.*

Proof. Without loss of generality, evaluate $\sum_{\substack{\mathbf{0} \leq \mathbf{y} \leq \mathbf{k} \\ B\mathbf{y} = \mathbf{b}}} x^{A\mathbf{y}}$ in all but the last variable, and let $q(x)$ be the resulting Laurent polynomial. It remains to check that $\{[x^\beta]q\}_{\beta \in \mathbb{Z}}$ is a log-concave sequence.

Let $\mathbf{a}_{(1)}, \mathbf{a}_{(2)}, \dots, \mathbf{a}_{(m)}$ be the rows of A , and suppose we are evaluating p at $y_1 = r_1, \dots, y_{m-1} = r_{m-1}$. Then:

$$\begin{aligned} q(x) &= \sum_{\substack{\mathbf{0} \leq \mathbf{y} \leq \mathbf{k} \\ B\mathbf{y} = \mathbf{b}}} r_1^{\mathbf{a}_{(1)}^T \mathbf{y}} \dots r_{m-1}^{\mathbf{a}_{(m-1)}^T \mathbf{y}} x^{\mathbf{a}_{(m)}^T \mathbf{y}} \\ &= \sum_{\substack{\mathbf{0} \leq \mathbf{y} \leq \mathbf{k} \\ B\mathbf{y} = \mathbf{b} \\ A\mathbf{y} = \mathbf{w}}} r_1^{w_1} \dots r_{m-1}^{w_{m-1}} x^{w_m} \end{aligned}$$

Since the set $\{\mathbf{y} : B\mathbf{y} = \mathbf{b}, \mathbf{0} \leq \mathbf{y} \leq \mathbf{k}\}$ is bounded, one can choose some integer vector γ such that $-\gamma \leq \mathbf{w} = A\mathbf{y} \leq \gamma$ for any $\mathbf{y}$ in this set. Then:

$$q(x) = \sum_{\beta \in \mathbb{Z}} \left(\sum r_1^{w_1} \dots r_{m-1}^{w_{m-1}} \right) x^\beta,$$

where the second sum is over the set:

$$T = \left\{ \begin{bmatrix} \mathbf{y} \\ \mathbf{w} \end{bmatrix} : \begin{bmatrix} B & \mathbf{0} \\ A & -\text{id} \end{bmatrix} \begin{bmatrix} \mathbf{y} \\ \mathbf{w} \end{bmatrix} = \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \end{bmatrix}, \quad w_m = \beta, \quad \begin{bmatrix} \mathbf{0} \\ -\gamma \end{bmatrix} \leq \begin{bmatrix} \mathbf{y} \\ \mathbf{w} \end{bmatrix} \leq \begin{bmatrix} \mathbf{k} \\ \gamma \end{bmatrix} \right\}$$

Let M be the matrix that is obtained by appending a row of $e_{n+m}^T = [0 \ \dots \ 0 \ 1]$ row to the end of $\begin{bmatrix} B & \mathbf{0} \\ A & -\text{id} \end{bmatrix}$, i.e. $M = \begin{bmatrix} B & \mathbf{0} \\ A & -\text{id} \\ e_{n+m}^T \end{bmatrix}$. By basic properties of TU matrices, M is TU. Then we can rewrite:

$$T = \left\{ \begin{bmatrix} \mathbf{y} \\ \mathbf{w} \end{bmatrix} : M \begin{bmatrix} \mathbf{y} \\ \mathbf{w} \end{bmatrix} = \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \\ \beta \end{bmatrix}, \quad \begin{bmatrix} \mathbf{0} \\ -\gamma \end{bmatrix} \leq \begin{bmatrix} \mathbf{y} \\ \mathbf{w} \end{bmatrix} \leq \begin{bmatrix} \mathbf{k} \\ \gamma \end{bmatrix} \right\} = B \begin{bmatrix} \mathbf{0} \\ -\gamma \end{bmatrix}, \begin{bmatrix} \mathbf{k} \\ \gamma \end{bmatrix} \left(M, \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \\ \beta \end{bmatrix} \right).$$

Further rewrite $q(x)$ as follows:

$$q(x) = \sum_{\beta \in \mathbb{Z}} T \left(M, \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \\ \beta \end{bmatrix}, \begin{bmatrix} \mathbf{0} \\ -\gamma \end{bmatrix}, \begin{bmatrix} \mathbf{k} \\ \gamma \end{bmatrix}; [1 \ 1 \ \dots \ 1 \ r_1 \ \dots \ r_{m-1} \ 1]^T \right) x^\beta.$$

So the coefficients of $q(x)$ form the following sequence:

$$\left\{ T \left(M, \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \\ \beta \end{bmatrix}, \begin{bmatrix} \mathbf{0} \\ -\gamma \end{bmatrix}, \begin{bmatrix} \mathbf{k} \\ \gamma \end{bmatrix}; [1 \ 1 \ \dots \ 1 \ r_1 \ \dots \ r_{m-1} \ 1]^T \right) \right\}_{\beta \in \mathbb{Z}}$$

Observe that this sequence does not have any internal zeros. This is because given any integer vectors β_1, β_2 , the set $\{\mathbf{d} : M\mathbf{y} = \mathbf{d} \text{ for some } \beta_1 \leq \mathbf{y} \leq \beta_2\}$ is convex, and if $\mathbf{d}$ is an integer vector in this set, then $\beta_1 \leq \mathbf{y} \leq \beta_2, M\mathbf{y} = \mathbf{d}$ has an integer solution by Theorem 3.4.

Thus, this sequence is log-concave due to Theorem 3.1 and because $2 \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \\ \beta \end{bmatrix} = \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \\ \beta - 1 \end{bmatrix} + \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \\ \beta + 1 \end{bmatrix}$. $\square$

Remark 3.9. Suppose M is a general unimodular matrix. To prove that $P_{M,\mathbf{k}}$ is VLC, we need to show that $\sum_{\substack{\mathbf{0} \leq \mathbf{y} \leq \mathbf{k} \\ B\mathbf{y} = \mathbf{b}}} \mathbf{x}^{A\mathbf{y}}$ is VLC for every partition of the rows of M into two submatrices A and B . However, the proof of Theorem 3.8 does not always for unimodular matrices since $\begin{bmatrix} B & 0 \\ A & \text{id} \\ e_{n+m}^T \end{bmatrix}$ does not necessarily remain unimodular.

Corollary 3.10. *By Theorem 3.7 and Theorem 3.8, the $\mathbf{k}$ -solution Laurent polynomial of any TU matrix is VLC for any $\mathbf{k} \geq \mathbf{0}$.*

4 Capacity bounds for lattice points of TU polytopes

With Theorem 3.10 at our disposal, we can use Theorem 2.2 to derive capacity bounds for $B_{(\mathbf{0},\mathbf{k})}(M, \mathbf{b})$ based on $\text{cap}_{\mathbf{b}}(P_{M,\mathbf{k}})$ and $\text{Deg}_i^{\mathbf{b}}(P_{M,\mathbf{k}})$. In this section, we will focus on finding explicit bounds for the set $\text{Deg}_i^{\mathbf{b}}(P_{M,\mathbf{k}})$ based on M , $\mathbf{b}$ and $\mathbf{k}$. We look at general TU matrices in Section 4.1 and find a simply exponential lower bound for $B(M, \mathbf{b}) := \#\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n : M\mathbf{y} = \mathbf{b}\}$. In Section 4.2, we look at two special classes of TU matrices and find another lower bound for $B_M(\mathbf{b})$.

First, let us find an alternate characterization for both $\min \text{Deg}_i^{\mathbf{b}}(P_{M,\mathbf{k}})$ and $\max \text{Deg}_i^{\mathbf{b}}(P_{M,\mathbf{k}})$ using LP strong duality. For any matrix $M \in \mathbb{R}^{m \times n}$, let $\mathbf{m}_{(i)}$ be the i -th row of M and let $M_{[i+1,m]}$ denote the submatrix of M consisting of the rows $i+1, \dots, m$. Further define $\mathbf{s}_i$ to be the sum of the first i rows of M . In particular, if $\mathbf{b} \in \mathbb{R}^m$ is a vector, then $\mathbf{b}_{[i+1,m]}$ is the restriction of $\mathbf{b}$ to its $i+1, \dots, m$ -th entries.

Proposition 4.1. *Let M be an $m \times n$ TU matrix and $\mathbf{k} \in \mathbb{Z}^n$. For any $i \in [m-1]$:*

$$\max \text{Deg}_i^{\mathbf{b}}(P_{M,\mathbf{k}}) = \min\{\mathbf{b}_{[i+1,m]}^T \mathbf{w} + \mathbf{k}^T \mathbf{z} : (M_{[i+1,m]})^T \mathbf{w} + \mathbf{z} \geq \mathbf{m}_{(i)}, \mathbf{z} \geq \mathbf{0}\},$$

Similarly:

$$\min \text{Deg}_i^{\mathbf{b}}(P_{M,\mathbf{k}}) = \max\{\mathbf{b}_{[i+1,m]}^T \mathbf{w} - \mathbf{k}^T \mathbf{z} : (M_{[i+1,m]})^T \mathbf{w} - \mathbf{z} \leq \mathbf{m}_{(i)}, \mathbf{z} \geq \mathbf{0}\}.$$

Proof. The set of degrees of y_i in $[y_{i+1}^{\alpha_{i+1}} \cdots y_m^{\alpha_m}] P_{M,\mathbf{0},\mathbf{k}}$ is as follows:

$$\text{Deg}_i^{\mathbf{b}}(P_{M,\mathbf{k}}) = \{\mathbf{m}_{(i)}^T \mathbf{x} : M_{[i+1,m]} \mathbf{y} = \mathbf{b}_{[i+1,m]}, \mathbf{0} \leq \mathbf{y} \leq \mathbf{k}, \mathbf{y} \in \mathbb{Z}^n\},$$

since M is a TU matrix, a tight upper bound for this set would be the solution to the LP problem:

$$\max\{\mathbf{m}_{(i)}^T \mathbf{y} : M_{[i+1,m]} \mathbf{y} = b_{[i+1,m]}, \mathbf{0} \leq \mathbf{y} \leq \mathbf{k}\}. \quad (\text{P})$$

The dual of this LP is:

$$\min\{\mathbf{b}_{[i+1,m]}^T \mathbf{w} + \mathbf{k}^T \mathbf{z} : (M_{[i+1,m]})^T \mathbf{w} + \mathbf{z} \geq \mathbf{m}_{(i)}, \mathbf{z} \geq \mathbf{0}\}. \quad (\text{D})$$

By strong LP duality, the objective value of (P) and (D) are equal, completing the first part of the proof. The argument for $\min \text{Deg}_i^b(P_{M,\mathbf{k}})$ is similar. $\square$

Computing the optimal values of the LPs in Theorem 4.1 for all $i \in [m-1]$ can be computationally costly. Instead, we evaluate the objective functions of these LPs at feasible points, getting weaker bounds for $\text{Deg}_i^b(P_{M,\mathbf{k}})$. We can use this looser bound to obtain a capacity lower bound for $B_{(\mathbf{0},\mathbf{k})}(M, \mathbf{b})$ by Theorem 2.2. We will look at two different sets of feasible points for these LPs. First, we will let $\mathbf{w} = \mathbf{0}$ and adjust $\mathbf{z}$ accordingly to find a feasible point. This way we get capacity lower bounds for any TU matrix M , which are stated in Section 4.1. Second, we will let $\mathbf{z} = \mathbf{0}$ and adjust $\mathbf{w}$ accordingly. In case a feasible $\mathbf{w}$ exists, we can find an alternate capacity lower bound for $B_{(\mathbf{0},\mathbf{k})}(M, \mathbf{b})$, see Section 4.2. In particular, we find a lower bound for $B_{(\mathbf{0},\mathbf{k})}(M, \mathbf{b})$ when M has non-negative partial row sums and recover a lower bound for the number of lattice points of a flow polytope already presented in [LM26] and [LMY26].

4.1 General capacity bounds for TU polytopes

As proven in Theorem 3.10, the $\mathbf{k}$ -solution Laurent polynomial of any TU matrix is VLC. Thus we can obtain a lower bound for $B_{(\mathbf{0},\mathbf{k})}(M, \mathbf{b})$ using Theorem 2.2:

Theorem 4.2. *For any $m \times n$ TU matrix M and any integer vectors $\mathbf{k} \in \mathbb{Z}^n$, $\mathbf{b} \in \mathbb{Z}^m$ we have:*

$$B_{(\mathbf{0},\mathbf{k})}(M, \mathbf{b}) \geq \text{cap}_{\mathbf{b}}(P_{M,\mathbf{k}}) \cdot \prod_{i=1}^m \max \left\{ \frac{|\alpha_i - b_i|^{\alpha_i - b_i}}{(1 + |\alpha_i - b_i|)^{1 + |\alpha_i - b_i|}}, \frac{|\gamma_i - b_i|^{\gamma_i - b_i}}{(1 + |\gamma_i - b_i|)^{1 + |\gamma_i - b_i|}} \right\},$$

where:

$$\alpha_i = - \sum_{j \in [n]: M_{i,j} = -1} k_j, \quad \gamma_i = \sum_{j \in [n]: M_{i,j} = 1} k_j \quad \forall i \in [m].$$

Proof. It follows from Theorem 2.2 that:

$$B_{(\mathbf{0},\mathbf{k})}(M, \mathbf{b}) \geq \text{cap}_{\mathbf{b}}(P_{M,\mathbf{k}}) \cdot \prod_{i=1}^m \frac{|\beta_i - b_i|^{\beta_i - b_i}}{(1 + |\beta_i - b_i|)^{1 + |\beta_i - b_i|}},$$

where β_i is either a lower bound or an upper bound $\text{Deg}_i^b(P_{M,\mathbf{k}})$. Recall that for any $r \in \mathbb{R}$:

$$(r)_+ = \begin{cases} r & \text{if } r > 0 \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad (r)_- = \begin{cases} r & \text{if } r < 0 \\ 0 & \text{otherwise} \end{cases},$$

and for any vector $\mathbf{v} \in \mathbb{R}^n$, both $(\mathbf{v})_+$, $(\mathbf{v})_-$ are applied coordinate-wise. It is easy to check that the pair $\mathbf{w} = \mathbf{0}, \mathbf{z} = (\mathbf{m}_{(i)})_+$ satisfy $(M_{[i+1, m]})^T \mathbf{w} + \mathbf{z} \geq \mathbf{m}_{(i)}, \mathbf{z} \geq \mathbf{0}$. It follows from Theorem 4.1 that γ_i is indeed an upper bound for $\text{Deg}_i^b(P_{M, \mathbf{k}})$ for any $i \in [m-1]$. Similarly, $\mathbf{w} = \mathbf{0}, \mathbf{z} = -(\mathbf{m}_{(i)})_-$ is a feasible point for $(M_{[i+1, m]})^T \mathbf{w} - \mathbf{z} \leq \mathbf{m}_{(i)}, \mathbf{z} \geq \mathbf{0}$, implying that α_i is a lower bound for $\text{Deg}_i^b(P_{M, \mathbf{k}})$ for any $i \in [m-1]$.

It is also easy to check that α_m, γ_m are bounds for $\text{Deg}_m^b(P_{M, \mathbf{k}})$, since:

$$-\sum_{j \in [n]: M_{m,j} = -1} k_j \leq \sum_{j \in [n]} M_{m,j} x_j \leq \sum_{j \in [n]: M_{m,j} = 1} k_j, \quad \forall \mathbf{0} \leq \mathbf{x} \leq \mathbf{k}$$

Hence proving the statement. $\square$

Let us assume that the fiber $\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n : M\mathbf{y} = \mathbf{b}\}$ is finite. Then there should exist some $\mathbf{k}$ such that any $\mathbf{y}$ in this fiber satisfies $\mathbf{y} \leq \mathbf{k}$. For such a $\mathbf{k}$, the bound of Theorem 4.2 becomes a bound for $B(M, \mathbf{b}) = \#\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n : M\mathbf{y} = \mathbf{b}\}$. Through the rest of this section, we will explore the properties of TU matrices M with finite fibers and prove capacity bounds for $B(M, \mathbf{b})$.

Lemma 4.3. *Suppose M is an $m \times n$ TU matrix and the fiber $\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n : M\mathbf{y} = \mathbf{b}\}$ is nonempty and finite for some integer vector $\mathbf{b}$. Then any point of this fiber satisfies $y_i \leq \|\mathbf{b}\|_1$ for any $i \in [n]$.*

Proof. In fact, we can prove that any point $\mathbf{y}$ of $P = \{\mathbf{y} : M\mathbf{y} = \mathbf{b}, \mathbf{y} \geq \mathbf{0}\}$ satisfies $y_i \leq \|\mathbf{b}\|_1$.

We know that the polyhedron P is bounded since M is TU and $\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n : M\mathbf{y} = \mathbf{b}\}$ is finite. So P is the convex hull of its vertices and by convexity, it suffices to show that any vertex $\mathbf{y}^*$ of $\{\mathbf{y} : M\mathbf{y} = \mathbf{b}, \mathbf{y} \geq \mathbf{0}\}$ satisfies $y_i \leq \|\mathbf{b}\|_1$.

Let $r = \text{rank}(M)$ and let $\mathbf{y}^*$ be a vertex of P . The columns of M indexed by $\text{supp}(\mathbf{y}^*) := \{i \in [n] : y_i^* \neq 0\}$ are linearly independent. Otherwise, there would exist a nonzero vector $\mathbf{v}$ supported on $\text{supp}(\mathbf{y}^*)$ with $M\mathbf{v} = \mathbf{0}$, so that $\mathbf{y}^* \pm \varepsilon \mathbf{v} \in P$ for sufficiently small $\varepsilon > 0$, contradicting the fact that $\mathbf{y}^*$ is a vertex.

Let B be a set of r linearly independent columns of M containing $\text{supp}(\mathbf{y}^*)$ (B is also known as the basis of vertex $\mathbf{y}^*$ in linear programming). Then $y_i = 0$ for $i \notin B$. Let $M_{:,B}$ be the submatrix of M containing the columns indexed by B . $M_{:,B}$ still has rank r , so choose a $R \subseteq [m]$ with $|R| = r$ such that the submatrix of $M_{:,B}$ containing the rows indexed by R is invertible. Let this $r \times r$ submatrix be $M_{R,B}$.

Restricting $M\mathbf{y}^* = \mathbf{b}$ to the rows in R and columns in B gives us that $M_{R,B}\mathbf{y}_B^* = \mathbf{b}_R$, thus $\mathbf{y}_B = M_{R,B}^{-1}\mathbf{b}_R$. Since M is TU, $M_{R,B}^{-1}$ is TU as well, and any entry of $M_{R,B}^{-1}$ belongs to $\{0, \pm 1\}$. Consequently, for each $i \in B$ we have:

$$y_i \leq \sum_{j \in B} |b_j| \leq \|\mathbf{b}\|_1.$$

The same bound holds for $i \notin B$ because $y_i = 0$. $\square$

Corollary 4.4. Suppose $M \in \{0, \pm 1\}^{m \times n}$ is a TU matrix. Let $r = \text{rank}(M)$ and $c = \|b\|_1$. Then:

$$B(M, \mathbf{b}) \geq \text{cap}_{\mathbf{b}}(P_{M, c, \mathbf{1}}) \cdot \prod_{i=1}^m \max \left\{ \frac{|\alpha_i - b_i|^{\alpha_i - b_i}}{(1 + |\alpha_i - b_i|)^{1 + |\alpha_i - b_i|}}, \frac{|\gamma_i - b_i|^{\gamma_i - b_i}}{(1 + |\gamma_i - b_i|)^{1 + |\gamma_i - b_i|}} \right\},$$

where

$$\alpha_i = -c \cdot (\# \text{ of } (-1)\text{s in row } i), \quad \gamma_i = c \cdot (\# \text{ of } (+1)\text{s in row } i) \quad \forall i \in [m].$$

Proof. The bound is trivial if $B(M, \mathbf{b})$ is not finite. Otherwise, by Theorem 4.3 $B(M, \mathbf{b}) = B_{0, c, \mathbf{1}}(M, \mathbf{b})$. The result follows from Theorem 4.2. $\square$

We can further simplify the lower bound of Theorem 4.4 to obtain a simply exponential lower bound for $B(M, \mathbf{b})$ for large enough integer vectors $\mathbf{b}$.

Corollary 4.5. Suppose $M \in \{0, \pm 1\}^{m \times n}$ is a TU matrix with finite fibers. Then:

$$B(M, \mathbf{b}) \geq \text{cap}_{\mathbf{b}}(P_{M, \|b\|_1, \mathbf{1}}) \cdot \exp(-m - m \log(1 + n\|\mathbf{b}\|_1))$$

Proof. By Theorem 4.4 with $c = \|b\|_1$:

$$B(M, \mathbf{b}) \geq \text{cap}_{\mathbf{b}}(P_{M, c, \mathbf{1}}) \cdot \prod_{i=1}^m \max \left\{ \frac{|\alpha_i - b_i|^{\alpha_i - b_i}}{(1 + |\alpha_i - b_i|)^{1 + |\alpha_i - b_i|}}, \frac{|\gamma_i - b_i|^{\gamma_i - b_i}}{(1 + |\gamma_i - b_i|)^{1 + |\gamma_i - b_i|}} \right\},$$

where:

$$\alpha_i = -c \cdot (\# \text{ of } (-1)\text{s in row } i), \quad \gamma_i = c \cdot (\# \text{ of } (+1)\text{s in row } i) \quad \forall i \in [m].$$

Note that $\gamma_i \leq nc$ and $-nc \leq \alpha_i$. Furthermore, $z \mapsto \frac{z^z}{(1+z)^{1+z}}$ is a decreasing function over $\mathbb{R}_{>0}$. Thus:

$$\begin{aligned} B(M, \mathbf{b}) &\geq \text{cap}_{\mathbf{b}}(P_{M, c, \mathbf{1}}) \cdot \prod_{i=1}^m \max \left\{ \frac{|\alpha_i - b_i|^{\alpha_i - b_i}}{(1 + |\alpha_i - b_i|)^{1 + |\alpha_i - b_i|}}, \frac{|\gamma_i - b_i|^{\gamma_i - b_i}}{(1 + |\gamma_i - b_i|)^{1 + |\gamma_i - b_i|}} \right\} \\ &\geq \text{cap}_{\mathbf{b}}(P_{M, c, \mathbf{1}}) \cdot \prod_{i=1}^m \frac{|\text{sgn}(b_i)nc - b_i|^{\text{sgn}(b_i)nc - b_i}}{(1 + |\text{sgn}(b_i)nc - b_i|)^{1 + |\text{sgn}(b_i)nc - b_i|}} \\ &\geq \text{cap}_{\mathbf{b}}(P_{M, c, \mathbf{1}}) \cdot \prod_{i=1}^m \frac{(n\|\mathbf{b}\|_1)^{n\|\mathbf{b}\|_1}}{(1 + n\|\mathbf{b}\|_1)^{1 + n\|\mathbf{b}\|_1}}, \end{aligned}$$

where $\text{sgn}(r) = 1$ if $r > 0$ and $\text{sgn}(r) = -1$ otherwise. Using the inequality $\frac{k^k}{(1+k)^{1+k}} \geq \frac{1}{e(1+k)}$ for $k > 0$ we obtain:

$$\begin{aligned} B(M, \mathbf{b}) &\geq \text{cap}_{\mathbf{b}}(P_{M, c, \mathbf{1}}) \cdot \prod_{i=1}^m \frac{(n\|\mathbf{b}\|_1)^{n\|\mathbf{b}\|_1}}{(1 + n\|\mathbf{b}\|_1)^{1 + n\|\mathbf{b}\|_1}} \\ &\geq \text{cap}_{\mathbf{b}}(P_{M, c, \mathbf{1}}) \cdot \prod_{i=1}^m \exp(-1 - \log(1 + n\|\mathbf{b}\|_1)) \\ &= \text{cap}_{\mathbf{b}}(P_{M, c, \mathbf{1}}) \cdot \exp(-m - m \log(1 + n\|\mathbf{b}\|_1)). \end{aligned}$$

$\square$

If $\|\mathbf{b}\|_1$ is reasonably large so that $m + m \log(1 + n\|\mathbf{b}\|_1) = O(\|\mathbf{b}\|_1)$, one could rewrite the bound of Theorem 4.5 as a simply exponential bound in $\|\mathbf{b}\|_1$. The following corollary establishes a bound for $\|\mathbf{b}\|_1$ that gives us such a simply exponential bound for $B(M, \mathbf{b})$.

Corollary 4.6. *Suppose m, n are positive integers satisfying $\max\{m, n\} \geq 3$. Given a TU matrix $M \in \{0, \pm 1\}^{m \times n}$ with finite fibers and some $\mathbf{b} \in \mathbb{Z}^m$ satisfying $\|\mathbf{b}\|_1 \geq 6m \cdot \log(\max\{m, n\})$, we have:*

$$B(M, \mathbf{b}) \geq \text{cap}_{\mathbf{b}}(P_{M, \|\mathbf{b}\|_1, \mathbf{1}}) \cdot \exp(-\|\mathbf{b}\|_1)$$

Proof. Apply Theorem 4.5 to M and $\mathbf{b}$. We just have to check that:

$$\|\mathbf{b}\|_1 \geq m + m \log(1 + n\|\mathbf{b}\|_1),$$

or the following stronger inequality:

$$\|\mathbf{b}\|_1 - m \log(\|\mathbf{b}\|_1) \geq m + m \log(2) + m \log(n)$$

The function $z \mapsto z - m \log(z)$ is increasing for $z > m$, therefore it suffices to check the above inequality for $\|\mathbf{b}\|_1 = 6m \cdot \log(q)$ where $q = \max\{m, n\}$. Now we only need to show that:

$$6 \log q \geq 1 + \log(2) + \log(q) + \log(6q \log q),$$

or equivalently that $5 \log q \geq 1 + \log(2) + \log(6q \log q)$. Exponentiate both hand sides of this inequality to get the equivalent statement $q^4 \geq 12e \log q$, which holds for any $q \geq 3$. $\square$

4.2 Special cases: non-negative partial row sums

In Section 4.1, we employed capacity bounds for VLC polynomials stated in Theorem 2.2 alongside the dual characterization of $\max \text{Deg}_i^{\mathbf{b}}(P_{M, \mathbf{k}})$ and $\text{Deg}_i^{\mathbf{b}}(P_{M, \mathbf{k}})$ to find capacity bounds for $B(M, \mathbf{b})$ when M is a general TU matrix. Specifically, we use feasible points with $\mathbf{w} = \mathbf{0}$ for the dual LPs of Theorem 4.1.

In this section, rather than placing all of the weight of the choice of the feasible points on $\mathbf{z}$ by taking $\mathbf{w} = \mathbf{0}$, we will instead let $\mathbf{z} = \mathbf{0} \in \mathbb{Z}^n$ and search for a feasible $\mathbf{w} \in \mathbb{R}^{m-i}$. This will not be always possible, however if that M has non-negative partial row sums and the sum of all of its rows is the $\mathbf{0}$ vector, such a $\mathbf{w}$ always exists. In this case, we obtain a lower bound for $B(M, \mathbf{b})$ stated in Theorem 4.12. We can slightly improve this bound when M is the incidence matrix of a directed acyclic graph to recover [LMY26, Theorem 1.1], see Theorem 4.13. Lastly, we will state a lower bound for $B(M, \mathbf{b})$ when M is a TU matrix with non-negative partial row sums in Theorem 4.14.

Theorem 4.7. *Suppose $M \in \{0, \pm 1\}^{m \times n}$ and is a TU matrix and $\mathbf{k} \in \mathbb{Z}^n$. Let $\mathbf{s}_i = \mathbf{m}_{(1)} + \dots + \mathbf{m}_{(i)}$ be the sum of the first i rows of M for $i \in [m]$. Assume that $\mathbf{s}_i \geq \mathbf{0}$ for any $i \in [m-1]$ and $\mathbf{s}_m = \mathbf{0}$. Then :*

$$B_{(\mathbf{0}, \mathbf{k})}(M, \mathbf{b}) \geq \text{cap}_{\mathbf{b}}(P_{M, \mathbf{k}}) \cdot \prod_{i=1}^{m-1} \frac{|\sigma_i|^{\|\sigma_i\|}}{(1 + |\sigma_i|)^{1 + \|\sigma_i\|}}.$$

where $\sigma_i = b_1 + \dots + b_i$ for $i \in [m]$.

Proof. If $\mathbf{b} \notin \text{supp}(P_{M,\mathbf{k}})$, then $\text{cap}_{\mathbf{b}}(P_{M,\mathbf{k}}) = 0$ by Theorem 2.4 and Theorem 2.5, and the statement is trivial. Otherwise if there exists $\mathbf{0} \leq \mathbf{y} \leq \mathbf{k}$ satisfying $M\mathbf{y} = \mathbf{b}$, we should have $\sigma_m = 0$ since $\mathbf{s}_m = \mathbf{0}$.

Let $i \in [m-1]$. Since $\mathbf{m}_{(i)} + \dots + \mathbf{m}_{(m)} = \mathbf{s}_m - \mathbf{s}_{i-1} \leq \mathbf{0}$, the vector $\mathbf{w} = -\mathbf{1} \in \mathbb{Z}^{m-i}$ satisfies:

$$(M_{[i+1,m]})^T \mathbf{w} \geq \mathbf{m}_{(i)}.$$

So $-\mathbf{b}_{[i+1,m]}^T \mathbf{1} = \sigma_i - \sigma_m = \sigma_i$ is an upper bound for the set $\text{Deg}_i^{\mathbf{b}}(P_{M,\mathbf{k}})$ by Theorem 4.1. Moreover, $\mathbf{m}_{(m)} = -\mathbf{s}_{(m-1)} \leq \mathbf{0}$, thus 0 is an upper bound for the set of degrees of x_m appearing in $P_{M,\mathbf{k}}(\mathbf{x})$. By Theorem 2.2, we get:

$$\begin{aligned} B_{(\mathbf{0},\mathbf{k})}(M,\mathbf{b}) &\geq \text{cap}_{\mathbf{b}}(P_{M,\mathbf{k}}) \cdot \frac{|-b_m|^{-b_m}}{(1+|-b_m|)^{1+|-b_m|}} \cdot \prod_{i=1}^{m-1} \frac{|\sigma_i - b_i|^{|\sigma_i - b_i|}}{(1+|\sigma_i - b_i|)^{1+|\sigma_i - b_i|}} \\ &= \text{cap}_{\mathbf{b}}(P_{M,\mathbf{k}}) \cdot \frac{|\sigma_{m-1}|^{\sigma_{m-1}}}{(1+|\sigma_{m-1}|)^{1+|\sigma_{m-1}|}} \cdot \prod_{i=1}^{m-2} \frac{|\sigma_i|^{\sigma_i}}{(1+|\sigma_i|)^{1+|\sigma_i|}} \\ &= \text{cap}_{\mathbf{b}}(P_{M,\mathbf{k}}) \cdot \prod_{i=1}^{m-1} \frac{|\sigma_i|^{\sigma_i}}{(1+|\sigma_i|)^{1+|\sigma_i|}}, \end{aligned}$$

proving the desired bound. $\square$

We can apply Theorem 4.7 with a large enough $\mathbf{k}$ to bound $B(M,\mathbf{b})$. In fact, we can take the limit of the bound of Theorem 4.7 as $\mathbf{k} \rightarrow \infty$, by which we mean that every entry of $\mathbf{k}$ tends to infinity. If the lower bound converges as $\mathbf{k} \rightarrow \infty$, we get a bound for $B(M,\mathbf{b})$.

Lemma 4.8. *Suppose M is an $m \times n$ TU matrix such that $\mathbf{s}_i \geq \mathbf{0}$ for any $i \in [m]$. If M does not have an all zeros column, then all the fibers $\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n : M\mathbf{y} = \mathbf{b}\}$ are finite.*

Proof. Suppose not, and let $\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n : M\mathbf{y} = \mathbf{b}\}$ be infinite. Then by Dickson's lemma, there exists two distinct integer vectors $\mathbf{z} \geq \mathbf{y}$ satisfying $M\mathbf{y} = M\mathbf{z} = \mathbf{b}$. Thus, there exists some nonzero vector $\mathbf{v} \in \mathbb{R}_{\geq 0}^n$ satisfying $M\mathbf{v} = \mathbf{0}$. Let $j \in [n]$ be an index such that $v_j > 0$. We have:

$$\begin{bmatrix} \mathbf{1}_i \\ \mathbf{0}_{m-i} \end{bmatrix}^T M\mathbf{v} = \mathbf{s}_{(i)}^T \mathbf{v} = 0, \quad \forall i \in [m]$$

where $\mathbf{1}_i \in \mathbb{Z}^i$ is the all ones vector and $\mathbf{0}_{m-i} \in \mathbb{Z}^{m-i}$ the all zeros vector. Note that $\mathbf{s}_{(i)}, \mathbf{v} \geq \mathbf{0}$, so we must have $\mathbf{s}_{(i),j} = 0$ for any $i \in [m]$. That is:

$$\mathbf{s}_{(1),j} = \mathbf{s}_{(2),j} = \dots = \mathbf{s}_{(m),j},$$

which means that the j -th column of M is a zero column, contradicting our assumption. $\square$

Similar to the terminology of [LMY26], for a Laurent series $f(\mathbf{x})$ with non-negative coefficients, we define its domain of convergence as $\Omega_f = \{\mathbf{x} > \mathbf{0} : f(\mathbf{x}) \text{ converges}\}$, and define its capacity as $\text{cap}_{\alpha}(f) = \inf_{\mathbf{x} \in \Omega_f} \frac{f(\mathbf{x})}{\mathbf{x}^{\alpha}}$. If we define $\mathbf{f}(\mathbf{x}) = \infty$ for any $\mathbf{x} \notin \Omega_f$, this definition of capacity agrees with the earlier definition $\text{cap}_{\alpha}(f) = \inf_{\mathbf{x} > \mathbf{0}} \frac{f(\mathbf{x})}{\mathbf{x}^{\alpha}}$.

Let $\mathbf{a}_1, \dots, \mathbf{a}_n$ be the columns of M and define:

$$P_M(\mathbf{y}) := \sum_{\mathbf{y} \geq \mathbf{0}} \mathbf{x}^{M\mathbf{y}} = \prod_{j=1}^n \sum_{\ell \geq 0} \mathbf{x}^{\ell \mathbf{a}_j}.$$

Suppose M is an $m \times n$ TU matrix and has finite fibers, i.e. $\{\mathbf{y} \in \mathbb{Z}_{\geq 0}^n : M\mathbf{y} = \mathbf{b}\}$ is finite for any $\mathbf{b} \in \mathbb{Z}^m$. This holds, for example, when M has non-negative partial row sums and no zero columns (see Theorem 4.8). Then P_M is a well defined Laurent series with the domain of convergence $\Omega = \{\mathbf{x} \in \mathbb{R}_{>0}^m : \mathbf{x}^{\mathbf{a}_j} < 1 \ \forall j \in [n]\}$. Moreover, Ω is a nonempty set. By Motzkin's transposition theorem [Gul10, Theorem 7.17], there exists some $\mathbf{w}$ such that $\mathbf{w}^T \mathbf{a}_j > 0$ for any $j \in [n]$ since M has finite fibers. It follows that $e^{-r\mathbf{w}} \in \Omega$ for any $r > 0$. More precisely, the domain of convergence of each $\sum_{\ell \geq 0} \mathbf{x}^{\ell \mathbf{a}_j}$ is open and contains $e^{-r\mathbf{w}}$.

Lemma 4.9 ([LMY26, Proposition 6.4]). *Suppose $f_1(\mathbf{x}), \dots, f_n(\mathbf{x})$ are all Laurent series with non-negative coefficients. Let $\Omega_i \neq \emptyset$ be the domain of convergence of f_i , and suppose $\cap_{i=1}^n \text{int } \Omega_i \neq \emptyset$. Then:*

$$\text{cap}_{\alpha} \left(\prod_{i=1}^n f_i \right) = \sup_{\alpha_{(1)} + \dots + \alpha_{(n)} = \alpha} \prod_{i=1}^n \text{cap}_{\alpha_{(i)}}(f_i).$$

In fact, after taking the natural logarithm of both hand sides, the two programs are Fenchel primal-dual.

In our case, this implies a particularly nice expression for the capacity of P_M . We can also obtain a conceptually similar expression for the capacity of $P_{M,\mathbf{k}}$, which we discuss below.

Corollary 4.10. *Fix an $m \times n$ TU matrix M with finite fibers and an integer vector $\mathbf{b} \in \mathbb{Z}^m$. Letting $\mathcal{B}(M, \mathbf{b}) = \{\mathbf{y} \in \mathbb{R}_{\geq 0}^n : M\mathbf{y} = \mathbf{b}\}$, we have*

$$\text{cap}_{\mathbf{b}}(P_M) = \sup_{\mathbf{s} \in \mathcal{B}(M, \mathbf{b})} \prod_{j=1}^n \frac{(s_j + 1)^{s_j + 1}}{s_j^{s_j}}.$$

Proof. Let $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{Z}^m$ be the columns of M . By Theorem 4.9 we can write:

$$\text{cap}_{\mathbf{b}}(P_M) = \sup_{\mathbf{b}_1 + \dots + \mathbf{b}_n = \mathbf{b}} \prod_{j=1}^n \text{cap}_{\mathbf{b}_j} \left(\sum_{\ell \geq 0} \mathbf{x}^{\ell \mathbf{a}_j} \right).$$

We can further restrict to the case where $\mathbf{b}_j$ is a non-negative multiple of $\mathbf{a}_j$ by Theorem 2.5, otherwise, $\text{cap}_{\mathbf{b}_j} \left(\sum_{\ell \geq 0} \mathbf{x}^{\ell \mathbf{a}_j} \right) = 0$. It follows that:

$$\text{cap}_{\mathbf{b}}(P_M) = \sup_{\mathbf{s} \in \mathcal{B}(M, \mathbf{b})} \prod_{j=1}^n \text{cap}_{s_j \mathbf{a}_j} \left(\sum_{\ell \geq 0} \mathbf{x}^{\ell \mathbf{a}_j} \right).$$

One can see that $\text{cap}_{s_j \mathbf{a}_j} \left(\sum_{\ell \geq 0} \mathbf{x}^{\ell \mathbf{a}_j} \right) = \text{cap}_{s_j} \left(\sum_{\ell \geq 0} x^\ell \right)$ by a change of variables. We further have $\text{cap}_{s_j} \left(\sum_{\ell \geq 0} x^\ell \right) = \frac{(s_j+1)^{s_j+1}}{s_j^{s_j}}$ by [LMY26, Example 4.16], completing the proof. $\square$

We note that there is a similar expression for Theorem 4.10 in the case of the polytope $\mathcal{B}_{\mathbf{0}, \mathbf{k}}(M, \mathbf{b}) = \{\mathbf{y} \in \mathbb{R}^n : M\mathbf{y} = \mathbf{b}, \mathbf{0} \leq \mathbf{y} \leq \mathbf{k}\}$. In that case, the capacity of $P_{M, \mathbf{k}}$ is the sup over $\mathbf{s} \in \mathcal{B}_{\mathbf{0}, \mathbf{k}}(M, \mathbf{b})$ of the product of $e^{\mathcal{H}_{\{0, \dots, k_j\}}(s_j)}$, where $\mathcal{H}_{\{0, \dots, k_j\}}(s_j)$ is the entropy of the entropy-maximizing distribution on $\{0, \dots, k_j\}$ with expectation s_j (Note that this is precisely what is given in Theorem 4.10, for the geometric distribution on $\mathbb{Z}_{\geq 0}$ with expectation s_j .)

Now we can use Theorem 4.9 to prove that capacity is continuous under limiting $\mathbf{k} \rightarrow \infty$.

Lemma 4.11. *Let M be an $m \times n$ TU matrix with finite fibers and fix $\mathbf{b} \in \mathbb{Z}^m$. Then:*

$$\lim_{\mathbf{k} \rightarrow \infty} \text{cap}_{\mathbf{b}}(P_{M, \mathbf{k}}) = \text{cap}_{\mathbf{b}} P_M.$$

Proof. Observe that $P_M(\mathbf{x}) \geq P_{M, \mathbf{k}_1}(\mathbf{x}) \geq P_{M, \mathbf{k}_2}(\mathbf{x})$ for any $\mathbf{x} > \mathbf{0}$ and $\mathbf{k}_1 \geq \mathbf{k}_2$ (for $\mathbf{x} \notin \Omega$, let $P_M(\mathbf{x}) = \infty$), so $\text{cap}_{\mathbf{b}}(P_M) \geq \text{cap}_{\mathbf{b}}(P_{M, \mathbf{k}}) \geq 0$ for any $\mathbf{k}$. Since $\text{cap}_{\mathbf{b}}(P_{M, \mathbf{k}})$ is monotone in $\mathbf{k}$, the limit $\lim_{\mathbf{k} \rightarrow \infty} \text{cap}_{\mathbf{b}}(P_{M, \mathbf{k}})$ exists and $\text{cap}_{\mathbf{b}}(P_M) \geq \lim_{\mathbf{k} \rightarrow \infty} \text{cap}_{\mathbf{b}}(P_{M, \mathbf{k}})$.

For the other direction of the proof, let $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{Z}^m$ be the columns of M . By Theorem 4.9 and Theorem 2.5:

$$\begin{aligned} \text{cap}_{\mathbf{b}}(P_{M, \mathbf{k}}) &= \sup_{\mathbf{s} \in \mathcal{B}(M, \mathbf{b})} \prod_{j=1}^n \text{cap}_{s_j \mathbf{a}_j} \left(\sum_{\ell=0}^{k_j} \mathbf{x}^{\ell \mathbf{a}_j} \right), \\ \text{cap}_{\mathbf{b}}(P_M) &= \sup_{\mathbf{s} \in \mathcal{B}(M, \mathbf{b})} \prod_{j=1}^n \text{cap}_{s_j \mathbf{a}_j} \left(\sum_{\ell \geq 0} \mathbf{x}^{\ell \mathbf{a}_j} \right). \end{aligned}$$

Note that in the expression for $\text{cap}_{\mathbf{b}}(P_{M, \mathbf{k}})$, we can take the supremum to be over $\mathcal{B}(M, \mathbf{b})$ instead of $\mathcal{B}_{\mathbf{0}, \mathbf{k}}(M, \mathbf{b})$ because for any $\mathbf{s} \in \mathcal{B}(M, \mathbf{b}) \setminus \mathcal{B}_{\mathbf{0}, \mathbf{k}}(M, \mathbf{b})$, the corresponding product of capacities on the right hand side is zero.

Similar to the proof of Theorem 4.10, write $\text{cap}_{s_j \mathbf{a}_j} \left(\sum_{\ell=0}^{k_j} \mathbf{x}^{\ell \mathbf{a}_j} \right) = \text{cap}_{s_j} \left(\sum_{\ell=0}^{k_j} x^\ell \right)$. Therefore, for any choice of $\mathbf{s} \geq \mathbf{0}$ such that $M\mathbf{s} = \mathbf{b}$ we have:

$$\text{cap}_{\mathbf{b}}(P_{M, \mathbf{k}}) \geq \prod_{j=1}^n \text{cap}_{s_j} \left(\sum_{\ell=0}^{k_j} x^\ell \right),$$

By [BLP23, Lemma 5.5], $\lim_{k_j \rightarrow \infty} \text{cap}_{s_j}(\sum_{\ell=0}^{k_j} x^\ell) = \text{cap}_{s_j}(\sum_{\ell \geq 0} x^\ell)$. Send $k_j \rightarrow \infty$ for all $j \in [n]$ and write:

$$\lim_{\mathbf{k} \rightarrow \infty} \text{cap}_{\mathbf{b}}(P_{M,\mathbf{k}}) \geq \prod_{j=1}^n \text{cap}_{s_j} \left(\sum_{\ell \geq 0} x^\ell \right),$$

Note that this is true for any choice of $\mathbf{s} \in \mathcal{B}(M, \mathbf{b})$, so:

$$\begin{aligned} \lim_{\mathbf{k} \rightarrow \infty} \text{cap}_{\mathbf{b}}(P_{M,\mathbf{k}}) &\geq \sup_{\mathbf{s} \in \mathcal{B}(M, \mathbf{b})} \prod_{j=1}^n \text{cap}_{s_j} \left(\sum_{\ell \geq 0} x^\ell \right) \\ &= \text{cap}_{\mathbf{b}}(P_M). \end{aligned}$$

□

Using Theorem 4.11, we conclude the following corollary:

Corollary 4.12. *Suppose $M \in \{0, \pm 1\}^{m \times n}$ is a TU matrix with no zero columns and $\mathbf{k} \in \mathbb{Z}^n$. Let $\mathbf{s}_i = \mathbf{m}_{(1)} + \dots + \mathbf{m}_{(i)}$ be the sum of the first i rows of M for $i \in [m]$. Assume that $\mathbf{s}_i \geq \mathbf{0}$ for any $i \in [m-1]$ and $\mathbf{s}_m = \mathbf{0}$. Then :*

$$B(M, \mathbf{b}) \geq \text{cap}_{\mathbf{b}}(P_M) \cdot \prod_{i=1}^{m-1} \frac{|\sigma_i|^{\sigma_i}}{(1 + |\sigma_i|)^{1 + |\sigma_i|}}.$$

where $\sigma_i = b_1 + \dots + b_i$ for $i \in [m-1]$.

We remark that we took a different approach in Section 4.1 when working with general TU matrices. This is because it is not clear whether the lower bound of Theorem 4.2 converges as $\mathbf{k} \rightarrow \infty$, or if it converges, whether it converges to a non-zero number.

We will explore two applications of Theorem 4.12. We will apply it to incidence matrices of directed acyclic graphs, and to TU matrices with non-negative partial row sums.

When M is the incidence matrix of a directed acyclic graph G , $B(M, \mathbf{b})$ is called the Kostant partition function for integer flows of G . We refer to [Hum08] for a more extensive study of these quantities. The following is not a direct corollary of Theorem 4.12, but follows the same steps with a slightly different choice of feasible points for Theorem 4.1.

Corollary 4.13 ([LMY26, Theorem 1.1], see also [LM26]). *Let $G = ([n], E)$ be a directed acyclic graph with edges directed from the larger to the smaller vertex. For each $i \in [n]$, let G_i be the undirected induced subgraph of G on vertices $[i, n]$, and denote by C_i the connected component of i in G_i . Then:*

$$B(M, \mathbf{b}) \geq \text{cap}_{\mathbf{b}}(P_M) \cdot \prod_{i=2}^n \frac{|\sigma_i|^{\sigma_i}}{(1 + |\sigma_i|)^{1 + |\sigma_i|}},$$

where $M \in \{0, \pm 1\}^{n \times |E|}$ is the incidence matrix of G and $\sigma_i = \sum_{j \in C_i} b_j$.

Proof. When G is a directed acyclic graph with edges being directed from the larger vertices to smaller vertices, its incidence graph M has non-negative partial row sums and the sum of all its rows is the zero vector. In fact, for each $i \in [n]$, since there are no edges going into C_i , we have $\sum_{j \in C_i} \mathbf{m}_{(j)} \leq \mathbf{0}$, that is:

$$-(M_{[i+1, n]})^T \mathbf{1}_{C_i \setminus \{i\}} = - \sum_{j \in C_i \setminus \{i\}} \mathbf{m}_{(j)} \geq \mathbf{m}_{(i)},$$

where $\mathbf{1}_{C_i \setminus \{i\}}$ is the indicator vector of the set $C_i \setminus \{i\}$. Similar to Theorem 4.7 and by Theorem 2.2, we get:

$$B_{(\mathbf{0}, \mathbf{k})}(M, \mathbf{b}) \geq \text{cap}_{\mathbf{b}}(P_{M, \mathbf{k}}) \cdot \prod_{i=2}^n \frac{|\sigma_i|^{\sigma_i}}{(1 + |\sigma_i|)^{1 + |\sigma_i|}},$$

where $\sigma_i = \sum_{j \in C_i} b_j$. Now send $\mathbf{k} \rightarrow \infty$, similar to the proof of Theorem 4.12. $\square$

Lastly, we find a capacity bound for the cardinalities of fibers of TU matrices with nonnegative partial row sums.

Corollary 4.14. *Suppose M is an $m \times n$ TU matrix satisfying $\mathbf{s}_i \geq \mathbf{0}$ for all $i \in [m]$ (e.g. M only has 0,1 entries). Assume further that M does not have an all zero column. Then:*

$$B_M(\mathbf{b}) \geq \text{cap}_{\mathbf{b}}(P_M) \cdot \left[\prod_{i=1}^{m-1} \frac{|\sigma_i|^{\sigma_i}}{(1 + |\sigma_i|)^{1 + |\sigma_i|}} \right]^2 \cdot \frac{|\sigma_m|^{\sigma_m}}{(1 + |\sigma_m|)^{1 + |\sigma_m|}}.$$

where $\sigma_i = b_1 + \dots + b_i$ for $i \in [m]$.

Proof. Let $A = \begin{bmatrix} M \\ -JM \end{bmatrix}$, where J has 1s in the anti-diagonal and is zero everywhere else (that is, for any matrix B with proper dimensions, JB reverses the order of the rows of B). A is TU and satisfies the assumptions of Theorem 4.12. So by this corollary:

$$B_A \left(\begin{bmatrix} \mathbf{b} \\ -J\mathbf{b} \end{bmatrix} \right) \geq \text{cap}_{(\mathbf{b}, -J\mathbf{b})}(P_A) \cdot \prod_{i=1}^{m-1} \left[\frac{|\sigma_i|^{\sigma_i}}{(1 + |\sigma_i|)^{1 + |\sigma_i|}} \right]^2 \cdot \frac{|\sigma_m|^{\sigma_m}}{(1 + |\sigma_m|)^{1 + |\sigma_m|}}.$$

Note that both P_A and P_M are well-defined Laurent series by Theorem 4.8. We know that $P_A(x_i : i \in [2m]) = P_M \left(\frac{x_i}{x_{2m+1-i}} : i \in [m] \right)$, therefore:

$$\text{cap}_{(\mathbf{b}, -J\mathbf{b})}(P_A) = \inf_{x_i > 0 \forall i \in [2m]} \frac{P_M \left(\frac{x_i}{x_{2m+1-i}} : i \in [m] \right)}{\prod_{i \in [m]} \left(\frac{x_i}{x_{2m+1-i}} \right)^{b_i}} = \text{cap}_{\mathbf{b}}(P_M).$$

Furthermore:

$$\# \left\{ \mathbf{y} \in \mathbb{Z}_{\geq 0}^n : A\mathbf{y} = \begin{bmatrix} \mathbf{b} \\ -J\mathbf{b} \end{bmatrix} \right\} = \# \{ \mathbf{y} \in \mathbb{Z}_{\geq 0}^n : M\mathbf{y} = \mathbf{b} \}.$$

We conclude:

$$B_M(\mathbf{b}) \geq \text{cap}_{\mathbf{b}}(P_M) \cdot \left[\prod_{i=1}^{m-1} \frac{|\sigma_i|^{|\sigma_i|}}{(1 + |\sigma_i|)^{1+|\sigma_i|}} \right]^2 \cdot \frac{|\sigma_m|^{|\sigma_m|}}{(1 + |\sigma_m|)^{1+|\sigma_m|}}.$$

□

5 Efficiency of Approximation

We prove the algorithmic assertion of Theorem 1.3 using the interiority and ellipsoid argument of [SV14] (see Sections 5,6 of the arXiv version). We verify the necessary bounds directly for our Laurent polynomials, rather than applying their statements for 0/1 configurations verbatim. Throughout this section we assume M is full row-rank, but this is without loss of generality since one can always remove linearly dependent rows of M without changing the lattice point counts or the value of the capacity (up to consistency of $\mathbf{b}$).

Let $M \in \mathbb{Z}^{m \times n}$ be TU with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$, let $\mathbf{b} \in \mathbb{Z}^m$ and $\mathbf{k} \in \mathbb{Z}_{\geq 0}^n$, and write

$$p(\mathbf{x}) = P_{M,\mathbf{k}}(\mathbf{x}) = \prod_{j=1}^n \sum_{\ell=0}^{k_j} x^{\ell \mathbf{a}_j}, \quad \mathcal{E} = \text{cap}_{\mathbf{b}}(p).$$

We show that a deterministic algorithm can approximate $\log \mathcal{E}$ to additive error 2^{-q} in time polynomial in

$$m, \quad n, \quad q = \log \frac{1}{\epsilon}, \quad \log(1 + \|\mathbf{b}\|_{\infty}), \quad \log(1 + \|\mathbf{k}\|_{\infty}),$$

as long as $\mathcal{E} \neq 0$. All inputs are binary encoded and running time is measured in bit operations.

Finally, for Theorem 1.3, set $K = \|\mathbf{b}\|_1$ and $k_j = K$ before performing any reductions. This gives the stated dependence on $\log(1 + \|\mathbf{b}\|_1)$.

Reduction to $\mathbf{b}$ in the interior. The Newton polytope of p is the zonotope $Z = M[\mathbf{0}, \mathbf{k}]$; i.e. the sum of the line segments defined by $k_j \mathbf{a}_j$ for $j \in [n]$.

Let F be the smallest face of Z containing $\mathbf{b}$. Minimizing and maximizing each coordinate over $Q = \{\mathbf{y} \in \mathbb{R}^n : M\mathbf{y} = \mathbf{b}, \mathbf{0} \leq \mathbf{y} \leq \mathbf{k}\}$ identifies the set of coordinates forced to equal either 0 and k_j . Thus we can remove these columns of $M, \mathbf{k}$ to obtain $M', \mathbf{k}'$, and up to translation the zonotope $M'[\mathbf{0}, \mathbf{k}']$ is equal to F . Restricting to F and the associated variables preserves the corresponding coefficient and the value of the capacity by a standard optimization argument. Additionally, $\mathbf{b}$ is in the relative interior of the Newton polytope of the new polynomial, up to the same translation between $M'[\mathbf{0}, \mathbf{k}']$ and F . Thus we may WLOG assume that $\mathbf{b}$ is in the interior of Z .

Finally, note that this removal of columns may decrease the rank of M . Thus, we again remove linearly dependent rows of M at this point to account for this.

Bounding box for the minimizer. By Theorem 5.1 (see below), each facet inequality of Z can be written with normal $\mathbf{u}$ as

$$\mathbf{u}^\top \mathbf{y} \leq h, \quad \mathbf{u} \in \{0, +1, -1\}^m, \quad h \in \mathbb{Z}.$$

Since $\mathbf{b}$ is in the interior of Z , its slack in each inequality is at least one. Consequently,

$$B_2\left(\mathbf{b}, \frac{1}{\sqrt{m}}\right) \subseteq Z.$$

Now set $\Phi(\mathbf{z}) = \log p(e^{\mathbf{z}}) - \mathbf{b}^\top \mathbf{z}$, noting that $\log \text{cap}_{\mathbf{b}}(p) = \inf_{\mathbf{z} \in \mathbb{R}^m} \Phi(\mathbf{z})$. Every nonzero coefficient of p is a positive integer, and so

$$\Phi(\mathbf{z}) \geq \max_{\alpha \in \mathbb{Z}} (\alpha - \mathbf{b})^\top \mathbf{z} \geq \frac{\|\mathbf{z}\|_2}{\sqrt{m}}.$$

Thus Φ attains its minimum on $\mathbb{R}^m$, and every minimizer $\mathbf{z}^*$ satisfies

$$\|\mathbf{z}^*\|_2 \leq \sqrt{m} \cdot \log p(\mathbf{1}) = \sqrt{m} \sum_{j=1}^n \log(k_j + 1).$$

In particular, we may optimize over $[-R, R]^m$, where

$$R = \sqrt{m} \sum_{j=1}^n \log(k_j + 1).$$

This is the interiority-based bounding argument of [SV14], with $\log p(\mathbf{1})$ replacing the logarithm of the number of configurations.

Lemma 5.1. *Let M be an $m \times n$ totally unimodular matrix of rank m with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$, fix $\mathbf{k} \in \mathbb{Z}_{>0}^n$, and define the zonotope $Z = M[\mathbf{0}, \mathbf{k}] = \sum_{j=1}^n \mathbf{a}_j[0, k_j]$. Then the normal vectors of the facets of Z can be chosen to have entries in $\{0, +1, -1\}$.*

Proof. Let F be some facet of Z with normal vector $\mathbf{u}$. By [McM71], every facet of the zonotope Z is parallel to a Minkowski sum of some subset of the segments $\{\mathbf{a}_1[0, k_1], \dots, \mathbf{a}_n[0, k_n]\}$. Thus F is parallel to $\text{span}\{\mathbf{a}_j : \mathbf{u}^\top \mathbf{a}_j = 0\}$. WLOG by reindexing let $\mathbf{a}_1, \dots, \mathbf{a}_{m-1}$ span the $(m-1)$ -dimensional space parallel to F , and let C be the $m \times (m-1)$ submatrix of the corresponding columns of M , so that $\mathbf{u} \in \ker C^\top$. Define $v_i = (-1)^{i+1} \det C_i$ for all $i \in [m]$, where C_i is the matrix C with row i missing. A straightforward computation shows that $\mathbf{v}^\top \mathbf{a}_j = 0$ for all $j \in [m-1]$ (duplicate a given column of C and take determinant), and thus $\pm \mathbf{v}$ can be chosen as the normal vector of F . Since M , and thus C , is TU, this implies the desired result. $\square$

Approximate evaluation. We show that, for rational $\mathbf{z}$ and $0 < \tau < 1$, the objective

$$\Phi(\mathbf{z}) = -\mathbf{b}^\top \mathbf{z} + \sum_{j=1}^n H_{k_j+1}(\mathbf{a}_j^\top \mathbf{z}), \quad H_N(t) = \log \sum_{r=0}^{N-1} e^{rt},$$

can be evaluated to additive error τ in time polynomial in the input length, the encoding length of $\mathbf{z}$, and $\log \frac{1}{\tau}$. The dot products are computed exactly using rational arithmetic. To evaluate H_N , use $H_N(0) = \log N$ and, for $t \neq 0$, the geometric-series identity

$$H_N(t) = (N-1) \max\{t, 0\} + \ell(N|t|) - \ell(|t|), \quad \ell(s) = \log(1 - e^{-s}).$$

This avoids forming exponentials with large positive arguments.

We describe the precision needed to evaluate either $\ell(s)$, where $s \in \{|t|, N|t|\}$, to error 2^{-q} . Let B be the binary encoding length of the nonzero rational number t . Since $s \geq |t| \geq 2^{-B}$, we have

$$u := 1 - e^{-s} \geq \frac{1}{2} \min\{s, 1\} \geq 2^{-B-1}.$$

If $s \geq q+3$, then

$$|\ell(s)| \leq 2e^{-s} \leq 2^{-q-2},$$

so returning zero suffices. Otherwise $0 < s < q+3$. Approximate e^{-s} to absolute error $\eta = 2^{-(B+q+4)}$ via Taylor expansion; note that s is of magnitude $O(q)$ and requires $O(B+q)$ fractional bits. Now subtract this approximation from 1 to obtain $\tilde{u}$, which implies $|\tilde{u} - u| \leq \eta \leq \frac{u}{2}$ and thus $\tilde{u} \geq \frac{u}{2}$. Therefore by the mean value theorem, for some c between u and $\tilde{u}$ we have

$$|\log \tilde{u} - \log u| \leq |\tilde{u} - u| \cdot (\log t)'(c) \leq \eta \cdot \frac{2}{u} \leq 2^{-q-2}.$$

Approximate $\log \tilde{u}$ to an additional error of 2^{-q-2} via Taylor expansion of $\log(1-t)$, by first scaling the argument $\tilde{u}$ to the interval $[\frac{1}{2}, 1]$. This gives the required approximation to $\ell(s)$.

The rational term $(N-1) \max\{t, 0\}$ has encoding length polynomial in $B + \log N$, and errors in the two logarithms add without amplification. Thus $H_N(t)$ is computable to error 2^{1-q} in time polynomial in $B, \log N, q$. Taking

$$q = \left\lceil \log_2 \frac{4n}{\tau} \right\rceil$$

bounds the error in each term of the sum defining $\Phi(\mathbf{z})$ by $\frac{\tau}{2n}$; the cases $t = 0$ are evaluated to the same or better accuracy. Exact rational summation produces a rational V satisfying $|V - \Phi(\mathbf{z})| \leq \frac{\tau}{2} < \tau$.

Approximate gradients. Approximate gradients then follow from finite differences, which we now show. First, since $\mathbf{b} \in M[\mathbf{0}, \mathbf{k}]$, let $\mathbf{y} \in [\mathbf{0}, \mathbf{k}]$ be such that $\sum_{i=1}^n y_i \mathbf{a}_i = \mathbf{b}$. For fixed k and t , the first and second derivatives of H_{k+1} are the mean and variance of the distribution μ supported on $\{0, \dots, k\}$ with probabilities $\mu_\ell \propto e^{\ell t}$. Since μ is supported on $\{0, \dots, k\}$, this implies

$$\|\nabla \Phi(\mathbf{z})\|_2 = \left\| \sum_{j=1}^n \mathbf{a}_j \left[H'_{k_j+1}(\mathbf{a}_j^\top \mathbf{z}) - y_j \right] \right\|_2 \leq L := \sqrt{m} \sum_{j=1}^n k_j.$$

and

$$0 \preceq \nabla^2 \Phi(\mathbf{z}) = \sum_{j=1}^n \mathbf{a}_j \mathbf{a}_j^\top H''_{k_j+1}(\mathbf{a}_j^\top \mathbf{z}) \preceq I \cdot m \sum_{j=1}^n k_j^2,$$

so that

$$\|\nabla^2 \Phi(\mathbf{z})\|_{\text{op}} \leq H := m \sum_{j=1}^n k_j^2.$$

For coordinatewise gradient error δ , evaluate $\Phi(\mathbf{z})$ and $\Phi(\mathbf{z} + h\mathbf{e}_i)$, so that

$$\left| \partial_{z_i} \Phi(\mathbf{z}) - \frac{\Phi(\mathbf{z} + h\mathbf{e}_i) - \Phi(\mathbf{z})}{h} \right| \leq \frac{Hh}{2}.$$

Approximating $\Phi(\mathbf{z})$ by V_0 and $\Phi(\mathbf{z} + h\mathbf{e}_i)$ by V_i within error τ (as described above) then implies $\frac{V_i - V_0}{h}$ is within error $\frac{2\tau}{h}$. This implies

$$\left| \partial_{z_i} \Phi(\mathbf{z}) - \frac{V_i - V_0}{h} \right| \leq \frac{Hh}{2} + \frac{2\tau}{h}$$

Taking $h = \frac{\delta}{4H}$ and $\tau = \frac{\delta h}{8} = \frac{\delta^2}{32H}$, this implies an overall error of $\frac{Hh}{2} + \frac{2\tau}{h} = \frac{\delta}{8} + \frac{\delta}{4} < \delta$. Note that H is polynomial in m and $\mathbf{k}$, and thus $\log \frac{1}{\tau}$ depends polynomially on $\log m$, $\log(1 + \|\mathbf{k}\|_1)$, and $\log \frac{1}{\delta}$.

The algorithm. We now apply the approximate-oracle ellipsoid implementation of Section 6.2 (of the arXiv version) of [SV14] (see also Theorem 2.13), using the preceding bounds on the search region and the Lipschitz constant L of Φ . The value and gradient oracles established above provide its required numerical approximations. The number of oracle calls and the required number of accuracy bits are polynomial in the input length and $\log \frac{1}{\epsilon}$. Since our oracles run in time polynomial in these quantities and the encoding length of the query point, the resulting algorithm finds $\mathbf{z}^\circ$ satisfying

$$\Phi(\mathbf{z}^\circ) \leq \min \Phi + \frac{\epsilon}{2}$$

in polynomial bit complexity. Evaluating $\Phi(\mathbf{z}^\circ)$ to additive error $\frac{\epsilon}{2}$ then gives a rational $\mathcal{A}$ with $|\mathcal{A} - \log \mathcal{E}| \leq \epsilon$.

We note that although the total running time stated in [SV14] depends polynomially on $\frac{1}{\epsilon}$, its ellipsoid iteration bound depends only polynomially on $\log \frac{1}{\epsilon}$ (see Section 6.2, immediately before Lemma 6.3). The worse dependence comes from the approximate-counting oracle used there. Our value and gradient oracles instead run in time polynomial in the number of requested accuracy bits, so the same optimization framework gives polynomial dependence on $\log \frac{1}{\epsilon}$ in our setting.

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